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CHEMICALLY AMPLIFIED POSITIVE RESIST COMPOSITION AND RESIST PATTERN FORMING PROCESS

Abstract

An onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure has solvent solubility and generates an acid with reduced diffusion. A chemically amplified positive resist composition comprising the onium salt as a photoacid generator and a polymer adapted to be decomposed under the action of acid to increase its solubility in alkaline developer is processed by lithography to form a pattern of rectangular profile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No. 2024-031894 filed in Japan on Mar. 4, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This invention relates to a chemically amplified positive resist composition and a resist pattern forming process.

BACKGROUND ART

[0003] Pattern formation to a smaller feature size is required to meet the recent demand for higher integration in integrated circuits. Acid-catalyzed chemically amplified resist compositions are most often used in forming resist patterns with a feature size of 0.2 μm or less. High-energy radiation such as UV, deep-UV or EB is used as the energy source for exposure of these resist compositions. In particular, the EB lithography, which is utilized as the ultra-fine microfabrication technique, is also indispensable in processing a photomask blank into a photomask for use in the fabrication of semiconductor devices. Resist compositions for use in the EB lithography include positive ones wherein exposed regions are dissolved away to form a pattern and negative ones wherein exposed regions are retained to form a pattern. Either one which is easier to use is chosen in accordance with the morphology of the necessary resist pattern.

[0004] In general, the EB lithography is by writing an image with EB, without using a mask. In the case of positive resist, those regions of a resist film other than the regions to be retained are successively irradiated with EB having a minute area. In the case of negative resist, those regions of a resist film to be retained are successively irradiated with EB. The operation of successively scanning all finely divided regions on the work surface takes a long time as compared with full wafer exposure through a photomask. To prevent any throughput decline, a resist film having a high sensitivity is required. Because of a long image writing time, it is likely that a difference arises between an initially imaged portion and a lately imaged portion. The stability with time of the exposed portion in vacuum is one of the important performance factors. One of the important applications of chemically amplified resist material resides in processing of photomask blanks. Some photomask blanks have a surface material that can have an impact on the pattern profile of the overlying chemically amplified resist film, for example, a layer of a chromium compound, typically chromium oxide deposited on a photomask substrate. For high resolution and profile retention after etching, it is one important performance factor to maintain the profile of a resist film pattern rectangular independent of the type of substrate.

[0005] Attempts were made to ameliorate resist sensitivity and pattern profile in a controlled way by properly selecting and combining components used in resist compositions and adjusting processing conditions. One outstanding problem is the diffusion of acid, which has a significant impact on the resolution of a chemically amplified resist film. In the processing of photomasks, it is required that the profile of the resist pattern resulting from exposure does not change depending on the time taken until PEB. The major cause for time-dependent changes is the diffusion of acid generated upon exposure. Since the problem of acid diffusion has large impacts on sensitivity and resolution not only in the photomask processing, but also in general resist compositions, many studies are made thereon.

[0006] Patent Documents 1 and 2 describe acid generators capable of generating bulky acids upon

exposure, for thereby controlling acid diffusion and reducing roughness. Since these acid generators are still insufficient to control acid diffusion, it is desired to have an acid generator with more controlled diffusion.

[0007] Patent Document 3 discloses a resist composition comprising a base polymer having bound thereto an acid generator capable of generating a sulfonic acid upon light exposure whereby acid diffusion is controlled. This approach of controlling acid diffusion by binding repeat units capable of generating acid upon exposure to a base polymer is effective in forming a pattern with reduced LER. However, the base polymer having bound therein repeat units capable of generating acid upon exposure encounters a problem with respect to its solubility in organic solvent, depending on the structure and proportion of the relevant units.

[0008] Polymers comprising a major proportion of aromatic structure having an acidic side chain, for example, polyhydroxystyrene are useful in resist materials for the KrF lithography. These polymers are not used in resist materials for the ArF lithography since they exhibit strong absorption at a wavelength of around 200 nm. These polymers, however, are expected to form useful resist materials for the EB and EUV lithography for forming patterns of smaller size than the processing limit of ArF lithography because they offer high etching resistance.

[0009] Often used as the base polymer in positive resist compositions for EB and EUV lithography is a polymer having an acidic functional group on phenol side chain masked with an acid labile group (or acid-decomposable protective group). Upon exposure to high-energy radiation, the acid labile group is deprotected by the catalysis of an acid generated from a photoacid generator so that the polymer may turn soluble in alkaline developer. Typical of the acid labile group are tertiary alkyl, tert-butoxycarbonyl, and acetal groups. The use of protective groups (e.g., acetal groups) requiring a relatively low level of activation energy for deprotection offers the advantage that a resist film having a high sensitivity is obtainable. However, if the diffusion of generated acid is not fully controlled, deprotection reaction can occur even in the unexposed region of the resist film, giving rise to problems like a degradation of LER and a lowering of in-plane uniformity (CDU) of pattern line width.

[0010] Patent Documents 4 and 5 describe photoacid generators capable of generating a non-fluorinated aromatic sulfonic acid having a plurality of bulky alkyl substituents. Since a plurality of alkyl substituents are introduced, the generated acid has a higher molecular weight, which is effective for suppressing acid diffusion. The control of acid diffusion is insufficient for the purpose of forming small-size patterns. There remains room for further improvement.

[0011] In forming small-size patterns by the EB lithography, there is the problem that the profile of a resist pattern is inversely tapered by the influence of back scattering resulting from reflection of EB to the substrate. The inversely tapered profile may permit the resist pattern to collapse during development. For meeting the demand for further miniaturization, it is desired to solve the outstanding problem and to have a photoacid generator capable of suppressing acid diffusion while correcting the pattern profile.

CITATION LIST

[0012] Patent Document 1: JP-A 2009-053518 [0013] Patent Document 2: JP-A 2010-100604

[0014] Patent Document 3: JP-A 2011-022564 [0015] Patent Document 4: JP 6248882 [0016]

Patent Document 5: JP-A 2019-202974

SUMMARY OF THE INVENTION

[0017] Resist compositions are recently demanded which are capable of forming not only line-and-space (LS), isolated line (IL) and isolated space (IS) patterns of satisfactory profile, but also hole patterns of satisfactory profile. The acid generator described in Patent Document 4 generates a bulky acid, indicating that acid diffusion is controlled to some extent. This acid generator, however, gives rise to a problem that when a small-size pattern is formed, the pattern takes an inversely tapered profile and collapses during development in alkaline developer.

[0018] An object of the invention is to provide a chemically amplified positive resist composition

comprising an onium salt capable of generating an acid with controlled diffusion, and a resist pattern forming process using the composition.

[0019] The inventors have found that an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure has satisfactory solvent solubility. When a resist film is formed from a positive resist composition comprising the onium salt as an acid generator and processed by lithography, a pattern with reduced LER and improved CDU is obtained. By virtue of proper dissolution inhibition, a pattern of rectangular profile is obtained.

[0020] In one aspect, the invention provides a chemically amplified positive resist composition comprising (A) a photoacid generator and (B) a base polymer containing a polymer adapted to be decomposed under the action of acid to increase its solubility in alkaline developer. The photoacid generator (A) is an onium salt having the formula (A).

##STR00001##

[0021] Herein R^{sup.1} to R^{sup.12} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, [0022] one of R^{sup.13} and R^{sup.14} is a group having a partial structure having the formula (a) and the other is hydrogen, halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, [0023] at least two of R^{sup.1} to R^{sup.14} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and [0024] Z^{sup.+} is an onium cation,

##STR00002## [0025] wherein m₁ is 0 or 1, m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1, m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1, m₂+m₃ is from 0 to 4 when m₁=0 and from 0 to 6 when m₁=1, [0026] R^{sup.F} is fluorine, a C_{sub.1}-C_{sub.6} fluorinated alkyl group, C_{sub.1}-C_{sub.6} fluorinated alkoxy group, or C_{sub.1}-C_{sub.6} fluorinated alkylthio group, [0027] R^{sup.15} is a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, a plurality of R^{sup.15} may bond together to form a ring with the carbon atoms to which they are attached when m₃ is 2 or more, [0028] L^{sup.A} and L^{sup.B} are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and [0029] X^{sup.L} is a single bond or a C_{sub.1}-C_{sub.40} hydrocarbylene group which may contain a heteroatom.

[0030] The polymer comprises repeat units having the formula (B1).

##STR00003##

[0031] Herein a₁ is 0 or 1, a₂ is 0, 1 or 2, a₃ is an integer meeting 0≤a₃≤5+2(a₂)-a₄, a₄ is 1, 2 or 3, [0032] R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl, [0033] R^{sup.21} is halogen, an optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyl group, optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyloxy group, or optionally halogenated C_{sub.2}-C_{sub.8} saturated hydrocarbylcarbonyloxy group, and [0034] A^{sup.1} is a single bond or C_{sub.1}-C_{sub.10} saturated hydrocarbylene group in which some —CH_{sub.2}— may be replaced by —O—.

[0035] In one preferred embodiment, the onium salt has the formula (A1):

##STR00004## [0036] wherein R^{sup.1} to R^{sup.13}, R^{sup.15}, L^{sup.A}, X^{sup.L}, R^{sup.F}, m₁ to m₃, and Z^{sup.+} are as defined above.

[0037] In a more preferred embodiment, the onium salt has the formula (A2):

##STR00005## [0038] wherein R^{sup.5}, R^{sup.10} to R^{sup.13}, R^{sup.15}, L^{sup.A}, X^{sup.L}, R^{sup.F}, m₁ to m₃, and Z^{sup.+} are as defined above, [0039] m₄ and m₅ are each independently an integer of 0 to 4, [0040] R^{sup.16} and R^{sup.17} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom; when m₄ is 2 or more, a plurality of R^{sup.16} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m₅ is 2 or more, a plurality of R^{sup.17} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

[0041] Preferably, Z.sup.+ is a sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2):

##STR00006## [0042] wherein R.sup.ct1 to R.sup.ct5 are each independently halogen or a C.sub.1-C.sub.30 hydrocarbyl group which may contain a heteroatom, R.sup.ct1 and R.sup.ct2 may bond together to form a ring with the sulfur atom to which they are attached.

[0043] In one preferred embodiment, the polymer further comprises repeat units having the formula (B2-1).

##STR00007##

[0044] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0045] b1 is 0 or 1, b2 is 0, 1 or 2, b3 is an integer meeting $0 \leq b3 \leq 5 + 2(b2) - b4$, b4 is 1, 2 or 3, b5 is 0 or 1, [0046] R.sup.31 is halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, [0047] A.sup.2 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—, [0048] X is an acid labile group when b4=1, and X is hydrogen or an acid labile group, at least one being an acid labile group, when b4=2 or 3.

[0049] In one preferred embodiment, the polymer further comprises repeat units having the formula (B2-2).

##STR00008##

[0050] Herein c1 is 0, 1 or 2, c2 is 0, 1 or 2, c3 is an integer of 0 to 5, c4 is 0, 1 or 2, [0051] R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0052] R.sup.32 and R.sup.33 are each independently a C.sub.1-C.sub.10 hydrocarbyl group which may contain a heteroatom, R.sup.32 and R.sup.33 may bond together to form a ring with the carbon atom to which they are attached, [0053] R.sup.34 is each independently fluorine, a C.sub.1-C.sub.5 fluorinated alkyl group or C.sub.1-C.sub.5 fluorinated alkoxy group, [0054] R.sup.35 is each independently a C.sub.1-C.sub.10 hydrocarbyl group which may contain a heteroatom, and [0055] A.sup.3 is a single bond, phenylene, naphthylene or *—C(=O)—O-A.sup.31, A.sup.31 is a C.sub.1-C.sub.20 aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, a phenylene group or a naphthylene group, * designates a point of attachment to the carbon atom in the backbone.

[0056] In one preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B3), repeat units having the formula (B4), and repeat units having the formula (B5).

##STR00009##

[0057] Herein d is an integer of 0 to 6, e is an integer of 0 to 4, f1 is 0 or 1, f2 is 0, 1 or 2, f3 is an integer of 0 to 5, [0058] R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0059] R.sup.41 and R.sup.42 are each independently hydroxy, halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, [0060] R.sup.43 is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.2-C.sub.20 saturated hydrocarbylcarbonyloxy group, C.sub.2-C.sub.20 saturated hydrocarbyloxyhydrocarbyl group, C.sub.2-C.sub.20 saturated hydrocarbylthiohydrocarbyl group, halogen, nitro, or cyano, R.sup.43 may also be hydroxy when f2=1 or 2, and [0061] A.sup.4 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—.

[0062] In one preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B6), repeat units having the formula (B7), repeat units having the formula (B8), and repeat units having the formula (B9).

##STR00010##

[0063] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0064] X.sup.1 is a single

bond or phenylene group, [0065] X^{sup.2} is *^{sup.1}—C(=O)—O—X^{sup.21}—, *^{sup.1}C(=O)—NH—X^{sup.21}—, or *^{sup.1}—O—X^{sup.21}, X^{sup.21} is a C_{sub.1}-C_{sub.6} aliphatic hydrocarbylene group, phenylene group or a divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, *^{sup.1} is a point of attachment to X^{sup.1}, [0066] X^{sup.3} is each independently a single bond, phenylene, naphthylene, or *^{sup.2}—C(=O)—O—X^{sup.31}—, X^{sup.31} is a C_{sub.1}-C_{sub.10} aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, or phenylene or naphthylene group, *^{sup.2} is a point of attachment to the carbon atom in the backbone, [0067] X^{sup.4} is each independently a single bond, *^{sup.3}—X^{sup.41}—C(=O)—O—, *^{sup.3}—C(=O)—NH—X^{sup.41}—, or *^{sup.3}—O—X^{sup.41}—, X^{sup.41} is a C_{sub.1}-C_{sub.20} hydrocarbylene group which may contain a heteroatom, *^{sup.3} is a point of attachment to X^{sup.3}, [0068] X^{sup.5} is each independently a single bond, *^{sup.4}—X^{sup.51}—C(=O)—O—, *^{sup.4}—C(=O)—NH—X^{sup.51}—, or *^{sup.4}—O—X^{sup.51}—, X^{sup.51} is a C_{sub.1}-C_{sub.20} hydrocarbylene group which may contain a heteroatom, *^{sup.4} is a point of attachment to X^{sup.4}, [0069] X^{sup.6} is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *^{sup.2}—C(=O)—O—X^{sup.61}—, *^{sup.2}—C(=O)—N(H)—X^{sup.61}—, or *^{sup.2}—O—X^{sup.61}—, —X^{sup.61} is a C_{sub.1}-C_{sub.6} aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, *^{sup.2} is a point of attachment to the carbon atom in the backbone, [0070] R^{sup.51} and R^{sup.52} are each independently a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, R^{sup.1} and R^{sup.52} may bond together to form a ring with the sulfur atom to which they are attached, [0071] L^{sup.1} is a single bond, ether bond, ester bond, carbonyl group, sulfonate ester bond, carbonate bond or carbamate bond, [0072] R_f^{sup.1} and R_f^{sup.2} are each independently fluorine or a C_{sub.1}-C_{sub.6} fluorinated saturated hydrocarbyl group, [0073] R_f^{sup.3} and R_f^{sup.4} are each independently hydrogen, fluorine, or a C_{sub.1}-C_{sub.6} fluorinated saturated hydrocarbyl group, [0074] R_f^{sup.5} and R_f^{sup.6} are each independently hydrogen, fluorine, or a C_{sub.1}-C_{sub.6} fluorinated saturated hydrocarbyl group, excluding that all R_f^{sup.5} and R_f^{sup.6} are hydrogen at the same time, [0075] M^{sup.-} is a non-nucleophilic counter ion, [0076] A^{sup.+} is an onium cation, and [0077] g₁ and g₂ are each independently an integer of 0 to 3.

[0078] In one preferred embodiment, repeat units having an aromatic ring structure account for at least 60 mol % of the overall repeat units of the polymer in the base polymer.

[0079] The resist composition may further comprise (C) an organic solvent.

[0080] The resist composition may further comprise (D) a fluorinated polymer comprising repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3) and repeat units having the formula (D4) and optionally repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6).

##STR00011##

[0081] Herein j₁ is an integer of 1 to 3, j₂ is an integer meeting 0 ≤ j₂ ≤ 5 + 2(j₃) - j₁, j₃ is 0 or 1, k is an integer of 1 to 3, [0082] R^{sup.B} is each independently hydrogen, fluorine, methyl or trifluoromethyl, [0083] R^{sup.C} is each independently hydrogen or methyl, [0084] R^{sup.101}, R^{sup.102}, R^{sup.104} and R^{sup.105} are each independently hydrogen or a C_{sub.1}-C_{sub.10} saturated hydrocarbyl group, [0085] R^{sup.103}, R^{sup.106}, R^{sup.107} and R^{sup.108} are each independently hydrogen, a C_{sub.1}-C_{sub.15} hydrocarbyl group, C_{sub.1}-C_{sub.15} fluorinated hydrocarbyl group, or acid labile group, and when R^{sup.103}, R^{sup.106}, R^{sup.107} and R^{sup.108} each are a hydrocarbyl or fluorinated hydrocarbyl group, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond, [0086] R^{sup.109} is hydrogen or a C_{sub.1}-C_{sub.5} straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-

carbon bond, [0087] R.sup.110 is a C.sub.1-C.sub.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, [0088] R.sup.111 is a C.sub.1-C.sub.20 saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine, and in which some constituent —CH.sub.2— may be replaced by an ester bond or ether bond, [0089] Z.sup.1 is a C.sub.1-C.sub.20 (k+1)-valent hydrocarbon group or C.sub.1-C.sub.20 (k+1)-valent fluorinated hydrocarbon group, [0090] Z.sup.2 is a single bond, *—C(=O)—O— or *—C(=O)—NH—, * designates a point of attachment to the carbon atom in the backbone, [0091] Z.sup.3 is a single bond, —O—, *—C(=O)=O—Z.sup.31—Z.sup.32— or *—C(=O)—NH—Z.sup.31—Z.sup.32, Z.sup.31 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group, Z.sup.32 is a single bond, ester bond, ether bond, or sulfonamide bond, and * designates a point of attachment to the carbon atom in the backbone.

[0092] The resist composition may further comprise (E) a quencher. Preferably, the photoacid generator (A) and the quencher (E) are present in a weight ratio of less than 3/1.

[0093] The resist composition may further comprise a photoacid generator other than the photoacid generator defined herein.

[0094] In another aspect, the invention provides a resist pattern forming process comprising the steps of: [0095] applying the chemically amplified positive resist composition defined herein onto a substrate to form a resist film thereon, [0096] exposing the resist film patternwise to high-energy radiation, and [0097] developing the exposed resist film in an alkaline developer.

[0098] Typically, the high-energy radiation is EUV of wavelength 3 to 15 nm or EB.

[0099] In one embodiment, the substrate has the outermost surface of a chromium-containing material.

[0100] Typically, the substrate is a photomask blank.

Advantageous Effects of Invention

[0101] When processed by the microfabrication technology, especially EB and EUV lithography processes, a chemically amplified positive resist composition comprising an onium salt having formula (A) as a photoacid generator can form a resist pattern having a very high resolution and reduced LER. A pattern of rectangular profile is obtainable by virtue of properly inhibited dissolution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0102] The only FIGURE, FIG. 1 is a diagram showing the .sup.1H-NMR/DMSO-d.sub.6 spectrum of onium salt PAG-1 synthesized in Synthesis Example 1-1.

DETAILED DESCRIPTION OF THE INVENTION

[0103] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. “Optional” or “optionally” means that the subsequently described event or circumstances may or may not occur, and that description includes instances where the event or circumstance occurs and instances where it does not. The notation (C_n-C_m) means a group containing from n to m carbon atoms per group. In chemical formulae, the broken line (---) and asterisk (*) each designate a point of attachment, namely valence bond. Me stands for methyl and Ac for acetyl. As used herein, the term “halogenated” refers to a halogen-substituted or halogen-containing compound or group. For example, “fluorinated” refers to a fluorine-substituted or fluorine-containing compound or group. The terms “group” and “moiety” are interchangeable.

[0104] The abbreviations and acronyms have the following meaning. [0105] PAG: photoacid generator [0106] Mw: weight average molecular weight [0107] Mn: number average molecular weight [0108] Mw/Mn: molecular weight distribution or dispersity [0109] GPC: gel permeation chromatography [0110] PEB: post-exposure baking [0111] LER: line edge roughness [0112] CDU:

critical dimension uniformity

[0113] It is understood that for some structures represented by chemical formulae, there can exist enantiomers and diastereomers because of the presence of asymmetric carbon atoms. In such a case, a single formula collectively represents all such isomers. The isomers may be used alone or in admixture.

[Chemically Amplified Positive Resist Composition]

[0114] One embodiment of the invention is a chemically amplified positive resist composition comprising (A) a photoacid generator in the form of an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure and (B) a base polymer containing a specific polymer.

(A) Photoacid Generator

[0115] The photoacid generator (A) is an onium salt having the formula (A).

##STR00012##

[0116] In formula (A), R^{sup.1} to R^{sup.12} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom. One of R^{sup.13} and R^{sup.14} is a group having a partial structure having the formula (a) shown later while the other is hydrogen, halogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom.

[0117] Suitable halogen atoms include fluorine, chlorine, bromine, and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C_{sub.1}-C_{sub.20} alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, heptadecyl, octadecyl, nonadecyl, and icosyl; C_{sub.3}-C_{sub.20} cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, and adamantyl; C_{sub.2}-C_{sub.20} alkenyl groups such as vinyl, allyl, propenyl, butenyl, and hexenyl; C_{sub.3}-C_{sub.20} cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C_{sub.6}-C_{sub.20} aryl groups such as phenyl and naphthyl; C_{sub.7}-C_{sub.20} aralkyl groups such as benzyl, 1-phenylethyl, and 2-phenylethyl, and combinations thereof. Inter alia, aryl groups are preferred. In the hydrocarbyl group, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, cyano, fluorine, chlorine, bromine, iodine, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0118] At least two of R^{sup.1} to R^{sup.14} may bond together to form a ring with the carbon to which they are attached, or the carbon to which they are attached and the intervenient carbon. Examples of the ring thus formed include aliphatic rings such as cyclopropane, cyclobutane, cyclopentane, cyclohexane, norbornane, and adamantane rings and aromatic rings such as benzene, naphthalene and anthracene rings. In the ring, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the ring may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Of the rings, aromatic rings are preferred, and benzene rings are more preferred.

[0119] In formula (A), one of R^{sup.13} and R^{sup.14} is a group having a partial structure having the following formula (a). It is preferred from the aspect of synthesis that R^{sup.14} be a group having a partial structure having the formula (a).

##STR00013##

[0120] In formula (a), m₁ is 0 or 1. The relevant structure is a benzene ring when m₁=0, and a

naphthalene ring when $m_1=1$. The benzene ring corresponding to $m_1=0$ is preferred from the aspect of solvent solubility. The subscript m_2 is an integer of 0 to 4 when $m_1=0$ and an integer of 0 to 6 when $m_1=1$. Preferably, m_2 is 4 when $m_1=0$, and m_2 is 4, 5 or 6 when $m_1=1$. The subscript m_3 is an integer of 0 to 3 when $m_1=0$ and an integer of 0 to 5 when $m_1=1$. The sum m_2+m_3 is an integer of 0 to 4 when $m_1=0$ and an integer of 0 to 6 when $m_1=1$.

[0121] In formula (a), $R_{sup.F}$ is fluorine, a C.sub.1-C.sub.6 fluorinated alkyl group, C.sub.1-C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group. $R_{sup.F}$ is preferably fluorine, trifluoromethyl, difluoromethyl, trifluoromethoxy, difluoromethoxy, trifluoromethylthio or difluoromethylthio, more preferably fluorine, trifluoromethyl or trifluoromethoxy. When fluorine or a fluorinated substituent group is contained, the acid strength of the generated acid is so enhanced due to the electron attractive effect that deprotection reaction of acid labile groups such as tertiary ester or tertiary ether groups may smoothly take place. When m_2 is 2, 3 or 4, a plurality of $R_{sup.F}$ may be identical or different.

[0122] In formula (a), $R_{sup.15}$ is a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl group $R_{sup.1}$, but not limited thereto. When m_3 is 2 or more, a plurality of $R_{sup.15}$ may be identical or different. When m_3 is 2 or more, a plurality of $R_{sup.15}$ may bond together to form a ring with the carbon atoms to which they are attached. The ring is preferably a 5 to 8-membered ring.

[0123] In formula (a), $L_{sup.A}$ and $L_{sup.B}$ are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond. Inter alia, a single bond, ether bond or ester bond is preferred.

[0124] In formula (a), $X_{sup.L}$ is a single bond or a C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom. The hydrocarbylene group may be straight, branched or cyclic and examples thereof include alkanediyl groups, cyclic saturated hydrocarbylene groups, and arylene groups. Suitable heteroatoms include oxygen, nitrogen and sulfur.

[0125] Examples of the C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom, represented by $X_{sup.L}$, are shown below, but not limited thereto. Herein, * designates a point of attachment to $L_{sup.A}$ or $L_{sup.B}$.

##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020##

[0126] Of these, $X_{sup.L_0}$ to $X_{sup.L_22}$ and $X_{sup.L_47}$ to $X_{sup.L_58}$ are preferred.

[0127] Of the onium salts having formula (A), those having the formula (A1) are preferred.

##STR00021##

[0128] Herein $R_{sup.1}$ to $R_{sup.13}$, $R_{sup.15}$, $L_{sup.A}$, $X_{sup.L}$, $R_{sup.F}$, m_1 to m_3 , and $Z_{sup.+}$ are as defined above.

[0129] Of the onium salts having formula (A1), those having the formula (A2) are more preferred.

##STR00022##

[0130] In formula (A2), $R_{sup.5}$, $R_{sup.10}$ to $R_{sup.13}$, $R_{sup.15}$, $L_{sup.A}$, $X_{sup.L}$, $R_{sup.F}$, m_1 to m_3 , and $Z_{sup.+}$ are as defined above.

[0131] In formula (A2), m_4 and m_5 are each independently an integer of 0 to 4. From the aspect of reactant availability, m_4 and m_5 each are preferably 0, 1 or 2.

[0132] In formula (A2), $R_{sup.16}$ and $R_{sup.17}$ are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. Examples of the halogen and hydrocarbyl group are as exemplified above for the halogen and hydrocarbyl group represented by $R_{sup.1}$ to $R_{sup.14}$.

[0133] When m_4 is 2 or more, a plurality of $R_{sup.16}$ may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon. When m_5 is 2 or more, a plurality of $R_{sup.17}$ may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient

carbon. Examples of the ring are as exemplified above for the ring that is formed by at least two of R.sup.1 to R.sup.14.

[0134] Examples of the anion in the onium salt having formula (A) are shown below, but not limited thereto.

##STR00023## ##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028##
##STR00029## ##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034##
##STR00035## ##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040##
##STR00041## ##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046##
##STR00047## ##STR00048## ##STR00049##
##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055##
##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061##
##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067##
##STR00068## ##STR00069##
##STR00070## ##STR00071## ##STR00072## ##STR00073## ##STR00074## ##STR00075##
##STR00076## ##STR00077## ##STR00078## ##STR00079## ##STR00080## ##STR00081##
##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087##
##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093##
##STR00094## ##STR00095## ##STR00096## ##STR00097##
##STR00098## ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103##
##STR00104## ##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109##
##STR00110## ##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115##
##STR00116## ##STR00117## ##STR00118## ##STR00119## ##STR00120## ##STR00121##
##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126## ##STR00127##
##STR00128## ##STR00129## ##STR00130##
##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136##
##STR00137## ##STR00138## ##STR00139## ##STR00140## ##STR00141## ##STR00142##
##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148##
##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154##
##STR00155## ##STR00156## ##STR00157##
##STR00158## ##STR00159## ##STR00160## ##STR00161## ##STR00162## ##STR00163##
##STR00164## ##STR00165## ##STR00166## ##STR00167## ##STR00168## ##STR00169##
##STR00170## ##STR00171## ##STR00172## ##STR00173## ##STR00174## ##STR00175##
##STR00176## ##STR00177## ##STR00178## ##STR00179## ##STR00180## ##STR00181##
##STR00182## ##STR00183## ##STR00184## ##STR00185## ##STR00186##
##STR00187## ##STR00188## ##STR00189## ##STR00190## ##STR00191## ##STR00192##
##STR00193## ##STR00194## ##STR00195## ##STR00196## ##STR00197## ##STR00198##
##STR00199## ##STR00200## ##STR00201## ##STR00202## ##STR00203## ##STR00204##
##STR00205##
##STR00206## ##STR00207## ##STR00208## ##STR00209## ##STR00210## ##STR00211##
##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217##
##STR00218## ##STR00219## ##STR00220## ##STR00221## ##STR00222## ##STR00223##
##STR00224## ##STR00225## ##STR00226## ##STR00227## ##STR00228## ##STR00229##
##STR00230## ##STR00231## ##STR00232## ##STR00233## ##STR00234##
##STR00235## ##STR00236## ##STR00237## ##STR00238## ##STR00239## ##STR00240##
##STR00241## ##STR00242## ##STR00243## ##STR00244## ##STR00245## ##STR00246##
##STR00247## ##STR00248## ##STR00249## ##STR00250## ##STR00251## ##STR00252##
##STR00253## ##STR00254## ##STR00255## ##STR00256## ##STR00257## ##STR00258##
##STR00259## ##STR00260## ##STR00261## ##STR00262## ##STR00263## ##STR00264##
##STR00265## ##STR00266## ##STR00267## ##STR00268## ##STR00269## ##STR00270##
##STR00271##

##STR00272## ##STR00273## ##STR00274## ##STR00275## ##STR00276## ##STR00277##
##STR00278## ##STR00279## ##STR00280## ##STR00281## ##STR00282## ##STR00283##
##STR00284## ##STR00285## ##STR00286## ##STR00287## ##STR00288## ##STR00289##
##STR00290## ##STR00291##

##STR00292## ##STR00293## ##STR00294## ##STR00295## ##STR00296## ##STR00297##
##STR00298## ##STR00299## ##STR00300## ##STR00301## ##STR00302## ##STR00303##
##STR00304## ##STR00305## ##STR00306## ##STR00307## ##STR00308## ##STR00309##
##STR00310## ##STR00311## ##STR00312## ##STR00313## ##STR00314## ##STR00315##
##STR00316## ##STR00317## ##STR00318## ##STR00319## ##STR00320## ##STR00321##
##STR00322## ##STR00323## ##STR00324## ##STR00325## ##STR00326## ##STR00327##
##STR00328## ##STR00329## ##STR00330##

##STR00331## ##STR00332## ##STR00333## ##STR00334## ##STR00335## ##STR00336##
##STR00337## ##STR00338## ##STR00339## ##STR00340## ##STR00341## ##STR00342##
##STR00343## ##STR00344## ##STR00345## ##STR00346## ##STR00347## ##STR00348##
##STR00349## ##STR00350##

##STR00351## ##STR00352## ##STR00353## ##STR00354## ##STR00355## ##STR00356##
##STR00357## ##STR00358## ##STR00359## ##STR00360##

[0135] In formula (A), Z^{sup.+} is an onium cation, preferably sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2).

##STR00361##

[0136] In formulae (cation-1) and (cation-2), R^{sup.ct1} to R^{sup.ct5} are each independently halogen or a C_{sub.1}-C_{sub.30} hydrocarbyl group which may contain a heteroatom. Suitable halogen atoms include fluorine, chlorine, bromine and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C_{sub.1}-C_{sub.30} alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl; C_{sub.3}-C_{sub.30} cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, adamantyl; C_{sub.2}-C_{sub.30} alkenyl groups such as vinyl, allyl, propenyl, butenyl, hexenyl; C_{sub.3}-C_{sub.30} cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C_{sub.6}-C_{sub.30} aryl groups such as phenyl, naphthyl, thienyl; C_{sub.7}-C_{sub.30} aralkyl groups such as benzyl, 1-phenylethyl, 2-phenylethyl, and combinations thereof. Inter alia, the aryl groups are preferred. In the hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some —CH_{sub.2}— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, nitro, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0137] Also, R^{sup.ct1} and R^{sup.ct2} may bond together to form a ring with the sulfur atom to which they are attached. Exemplary structures of the ring are shown below.

##STR00362##

[0138] The broken line designates a point of attachment to R^{sup.ct3}.

[0139] Examples of the sulfonium cation having formula (cation-1) are shown below, but not limited thereto.

##STR00363## ##STR00364## ##STR00365## ##STR00366## ##STR00367## ##STR00368##
##STR00369## ##STR00370## ##STR00371## ##STR00372## ##STR00373## ##STR00374##
##STR00375## ##STR00376## ##STR00377## ##STR00378## ##STR00379## ##STR00380##
##STR00381## ##STR00382##

##STR00383## ##STR00384## ##STR00385## ##STR00386## ##STR00387## ##STR00388##
##STR00389## ##STR00390## ##STR00391## ##STR00392## ##STR00393## ##STR00394##
##STR00395## ##STR00396## ##STR00397## ##STR00398## ##STR00399##

##STR00400## ##STR00401## ##STR00402## ##STR00403## ##STR00404## ##STR00405##
##STR00406## ##STR00407## ##STR00408## ##STR00409## ##STR00410## ##STR00411##
##STR00412## ##STR00413## ##STR00414## ##STR00415## ##STR00416##

[0140] Examples of the iodonium cation having formula (cation-2) are shown below, but not limited thereto.

##STR00417## ##STR00418##

[0141] Examples of the onium salt include arbitrary combinations of anions with cations, both as exemplified above.

[0142] The onium salt can be synthesized by well-known methods. For example, a method of preparing an onium salt having the formula (PAG-1-ex) is described.

##STR00419##

[0143] Herein R.sup.1 to R.sup.13, R.sup.F, R.sup.15, L.sup.A, X.sup.L, m1 to m3, and Z.sup.+ are as defined above, M.sup.+ is a counter cation, and X.sup.- is a counter anion.

[0144] The first step is to react reactant SM-1 with reactant SM-2, which are commercially available or can be synthesized by any well-known synthesis method, to form an Intermediate In-1. A variety of condensing agents may be used when an ester bond is directly formed from the carboxy group on reactant SM-1 and the hydroxy group on reactant SM-2. Examples of the condensing agent used herein include N,N'-dicyclohexylcarbodiimide, N,N'-diisopropylcarbodiimide, 1-[3-(dimethylamino)propyl]-3-ethylcarbodiimide, and 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride. From the standpoint of ease of removal of urea compounds formed as by-products after reaction, it is desirable to use 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride. The reaction is carried out by dissolving reactants SM-1 and SM-2 in a halogenated solvent such as methylene chloride and adding a condensing agent. The reaction rate can be increased by adding 4-dimethylaminopyridine (DMAP) as a catalyst. While it is desirable in view of yield to monitor the reaction by silica gel thin layer chromatography (TLC) until the reaction is complete, the reaction time is typically about 12 to 24 hours. After the reaction is terminated, urea compounds formed by side reactions are removed by filtration or water-washing if necessary. The reaction solution is subjected to standard aqueous work-up, obtaining Intermediate In-1. If necessary, the intermediate is purified by a standard technique such as chromatography or recrystallization.

[0145] The second step is a salt exchange between Intermediate In-1 and reactant SM-3 in the form of an onium salt: Z.sup.+X.sup.- to synthesize an onium salt PAG-1-ex. X.sup.- is preferably a chloride ion, bromide ion, iodide ion or methylsulfate anion because the exchange reaction readily takes place in a quantitative way. It is desirable in view of yield to monitor the reaction process by TLC. The reaction mixture is subjected to standard aqueous work-up, obtaining an onium salt PAG-1-ex. If necessary, the onium salt is purified by a standard technique such as chromatography or recrystallization.

[0146] In the above-illustrated scheme, the second step of ion exchange may be readily carried out by any well-known procedure, for example, with reference to JP-A 2007-145797.

[0147] It is noted that the preparation method according to the above scheme is merely exemplary and the method of preparing the inventive onium salt is not limited thereto.

[0148] The inventive onium salt is structurally characterized in that a substituted fused ring structure and an aromatic sulfonic acid structure are introduced in the anion. The substituted fused ring structure has a large excluded volume and thus serves as a bulky substituent to effectively inhibit the generated acid from diffusing. The substituted fused ring structure is also enough resistant to alkaline developer to prevent the pattern in unexposed region from losing a film thickness. The aromatic sulfonic acid structure, on the other hand, generates an acid of robust structure, which is effective for inhibiting acid diffusion. It is preferred that the aromatic ring that forms the aromatic sulfonic acid structure contain fluorine or an electron attractive sulfonate ester bond as a linking group. Then the generated acid has a higher acidity enough to efficiently

deprotect the acid labile groups in the base polymer. Since fluorine has a high EUV absorption effect, which is not so high as iodine, the inclusion of more fluorine atoms leads to a greater number of secondary electrons, which promotes decomposition of cations and contributes to a sensitivity increase. JP 7109178 proposes an alkanesulfonic acid type photoacid generator having 2 to 4 fluorine atoms, which has the problems of noticeable acid diffusion because of alkanesulfonic acid and poor solvent solubility which leaves the risk of development defects. By virtue of the synergy of these effects, the resist composition comprising the onium salt can form a resist pattern having a high sensitivity, low acid diffusion, and improved LWR of line patterns or improved CDU of hole patterns. The pattern is fully resistant to collapse. The resist composition is effective for forming small-size patterns.

[0149] In the chemically amplified positive resist composition, the amount of the photoacid generator (A) is preferably 0.1 to 20 parts by weight, more preferably 1 to 10 parts by weight per 80 parts by weight of a base polymer to be described just below. As long as the amount of photoacid generator (A) is in the range, an acid is generated in an amount necessary for deprotection of acid labile groups so that a pattern of satisfactory profile may be formed, and the composition has storage stability. The photoacid generator (A) may be used alone or in admixture of two or more.

(B) Base Polymer

[0150] The resist composition further comprises (B) a base polymer containing a polymer which is decomposed under the action of acid to increase its solubility in alkaline developer.

[0151] The polymer contains repeat units having the formula (B1), which are also referred to as repeat units B1.

##STR00420##

[0152] In formula (B1), a_1 is 0 or 1. The subscript a_2 is an integer of 0 to 2. The relevant structure is a benzene ring in case of $a_2=0$, a naphthalene ring in case of $a_2=1$, and an anthracene ring in case of $a_2=2$. The subscript a_3 is an integer meeting $0 \leq a_3 \leq 5+2(a_2)-a_4$, and a_4 is an integer of 1 to 3. When a_2 is 0, preferably a_3 is 0, 1, 2 or 3 and a_4 is 1, 2 or 3. When a_2 is 1 or 2, preferably a_3 is 0, 1, 2, 3 or 4 and a_4 is 1, 2 or 3.

[0153] In formula (B1), R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl.

[0154] In formula (B1), R^{sup.21} is halogen, an optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyl group, optionally halogenated C_{sub.1}-C_{sub.6} saturated hydrocarbyloxy group or optionally halogenated C_{sub.2}-C_{sub.8} saturated hydrocarbylcarbonyloxy group. The saturated hydrocarbyl group and saturated hydrocarbyl moiety in the saturated hydrocarbyloxy and saturated hydrocarbylcarbonyloxy groups may be straight, branched or cyclic. Examples thereof include alkyl groups such as methyl, ethyl, n-propyl, isopropyl, butyl, pentyl, hexyl, and structural isomers thereof; cycloalkyl groups such as cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl; and combinations thereof. A carbon count within the upper limit ensures a satisfactory solubility in alkaline developer. A plurality of R^{sup.21} may be identical or different when a_3 is 2 or more.

[0155] In formula (B1), A^{sup.1} is a single bond or C_{sub.1}-C_{sub.10} saturated hydrocarbylene group in which some constituent —CH_{sub.2}— may be replaced by —O—. The saturated hydrocarbylene group may be straight, branched or cyclic and examples thereof include C_{sub.1}-C_{sub.10} alkanediyl groups such as methylene, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, and structural isomers thereof; C_{sub.3}-C_{sub.10} cyclic saturated hydrocarbylene groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanedyl, and cyclohexanedyl; and combinations thereof. For the saturated hydrocarbylene group containing an ether bond, in case of $a_1=1$ in formula (B1), the ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ester oxygen. In case of $a_1=0$, the atom bonding to the backbone becomes an ether oxygen atom, and a second ether bond may be incorporated at any position excluding the position between the α - and β -carbons relative to the ether oxygen. Saturated hydrocarbylene groups having no more than 10 carbon atoms are desirable

because of a sufficient solubility in alkaline developer.

[0156] Preferred examples of the repeat units B1 wherein $a_1=0$ and A.sup.1 is a single bond (meaning that the aromatic ring is directly bonded to the backbone of the polymer), that is, repeat units free of a linker: —C(=O)—O-A.sup.1- include units derived from 3-hydroxystyrene, 4-hydroxystyrene, 5-hydroxy-2-vinylnaphthalene, and 6-hydroxy-2-vinylnaphthalene. Exemplary units are shown below, but not limited thereto.

##STR00421## ##STR00422## ##STR00423## ##STR00424## ##STR00425## ##STR00426##

[0157] Preferred examples of the repeat units B1 wherein $a_1=1$, that is, having a linker: —C(=O)—O-A.sup.1- are shown below, but not limited thereto. Herein R.sup.A is as defined above.

##STR00427## ##STR00428## ##STR00429## ##STR00430## ##STR00431## ##STR00432## ##STR00433## ##STR00434## ##STR00435##

[0158] The content of repeat units B1 is preferably 15 to 90 mol %, more preferably 15 to 80 mol % of the overall units of the polymer. When the polymer further comprises repeat units of at least one type selected from repeat units having formula (B3) and repeat units having formula (B4), which provide the polymer with higher etch resistance, the repeat units containing a phenolic hydroxy group as a substituent, the total content of repeat units B1 and repeat units B3 and/or B4 should preferably fall in the range. The repeat units B1 may be of one type or a combination of plural types.

[0159] In a preferred embodiment, the polymer further contains repeat units B2 having an acidic functional group protected with an acid labile group (i.e., repeat units protected with an acid labile group and adapted to turn alkali soluble under the action of acid) in order that the positive resist composition in an exposed region turn soluble in alkaline developer.

[0160] Typical of the repeat unit B2 is a unit having the formula (B2-1), also referred to as repeat unit B2-1.

##STR00436##

[0161] In formula (B2-1), b_1 is 0 or 1. The subscript b_2 is an integer of 0 to 2. The structure represents a benzene skeleton when $b_2=0$, a naphthalene skeleton when $b_2=1$, and an anthracene skeleton when $b_2=2$. The subscript b_3 is an integer meeting $0 \leq b_3 \leq 5+2(b_2)-b_4$. The subscript b_4 is an integer of 1 to 3, and b_5 is 0 or 1. When $b_2=0$, preferably b_3 is 0, 1, 2 or 3 and b_4 is 1, 2 or 3. When $b_2=1$ or 2, preferably b_3 is 0, 1, 2, 3 or 4 and b_4 is 1, 2 or 3.

[0162] In formula (B2-1), R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl.

[0163] In formula (B2-1), R.sup.31 is halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group. The saturated hydrocarbyl group and saturated hydrocarbyl moiety in the saturated hydrocarbyloxy group and saturated hydrocarbylcarbonyloxy group may be straight, branched or cyclic, and examples thereof include alkyl groups such as methyl, ethyl, n-propyl, isopropyl, butyl, pentyl, hexyl, and structural isomers thereof, cycloalkyl groups such as cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl, and combinations thereof. A carbon count within the upper limit ensures good solubility in alkaline developer. A plurality of R.sup.31 may be identical or different when b_3 is 2 or more.

[0164] In formula (B2-1), A.sup.2 is a single bond or a C.sub.1-C.sub.10 saturated hydrocarbylene group in which some constituent —CH.sub.2— may be replaced by —O—. The saturated hydrocarbylene group may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.10 alkanediyl groups such as methylene, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, and structural isomers thereof; C.sub.3-C.sub.10 cyclic saturated hydrocarbylene groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanediiyl, and cyclohexanediiyl; and combinations thereof. For the saturated hydrocarbylene group containing an ether bond, in case of $b_1=1$ in formula (B2-1), the ether bond may be incorporated at any position excluding the position between the α -carbon and β -carbon relative to the ester oxygen. In case of $b_1=0$, the atom that bonds with the backbone becomes an ethereal oxygen, and a second ether bond

may be incorporated at any position excluding the position between the α -carbon and β -carbon relative to that etheral oxygen. Saturated hydrocarbylene groups having no more than 10 carbon atoms are desirable because of a sufficient solubility in alkaline developer.

[0165] In formula (B2-1), X is an acid labile group when $b_4=1$, and hydrogen or an acid labile group, at least one X being an acid labile group, when $b_4=2$ or 3. That is, repeat units B2-1 have phenolic hydroxy groups bonded to an aromatic ring, at least one of which is protected with an acid labile group, or repeat units B2-1 have a carboxy group bonded to an aromatic ring, which is protected with an acid labile group. The acid labile group used herein is not particularly limited as long as it is commonly used in a number of well-known chemically amplified resist compositions and eliminated under the action of acid to release an acidic group.

[0166] Typical of the acid labile group is a tertiary saturated hydrocarbyl group. The tertiary saturated hydrocarbyl group is preferably of 4 to 18 carbon atoms because a monomer for use in polymerization is recoverable by distillation.

[0167] The saturated hydrocarbyl group bonded to the tertiary carbon atom in the tertiary saturated hydrocarbyl group is preferably of 1 to 15 carbon atoms. The C.sub.1-C.sub.15 saturated hydrocarbyl group may be straight, branched or cyclic and contain an oxygen-containing functional group such as an ether bond or carbonyl group in its carbon-carbon bond. The saturated hydrocarbyl groups bonded to the tertiary carbon atom may bond together to form a ring with the tertiary carbon atom to which they are attached.

[0168] Examples of the alkyl substituent include methyl, ethyl, propyl, adamantyl, norbornyl, tetrahydrofuran-2-yl, 7-oxanorbornan-2-yl, cyclopentyl, 2-tetrahydrofuryl, tricyclo[5.2.1.0.sup.2,6]decyl, 8-ethyl-8-tricyclo[5.2.1.0.sup.2,6]decyl, 3-methyl-3-tetracyclo[4.4.0.1.sup.2,5.1.sup.7,10]dodecyl, tetracyclo[4.4.0.1.sup.2,5.1.sup.7,10]dodecyl, and 3-oxo-1-cyclohexyl.

[0169] Examples of the tertiary saturated hydrocarbyl group include, but are not limited to, tert-butyl, tert-pentyl, 1-ethyl-1-methylpropyl, 1,1-diethylpropyl, 1,1,2-trimethylpropyl, 1-adamantyl-1-methylethyl, 1-methyl-1-(2-norbornyl)ethyl, 1-methyl-1-(tetrahydrofuran-2-yl)ethyl, 1-methyl-1-(7-oxanorbornan-2-yl)ethyl, 1-methylcyclopentyl, 1-ethylcyclopentyl, 1-propylcyclopentyl, 1-cyclopentylcyclopentyl, 1-cyclohexylcyclopentyl, 1-(2-tetrahydrofuryl)cyclopentyl, 1-(7-oxanorbornan-2-yl)cyclopentyl, 1-methylcyclohexyl, 1-ethylcyclohexyl, 1-cyclopentylcyclohexyl, 1-cyclohexylcyclohexyl, 2-methyl-2-norbornyl, 2-ethyl-2-norbornyl, 8-methyl-8-tricyclo[5.2.1.0.sup.2,6]decyl, 8-ethyl-8-tricyclo[5.2.1.0.sup.2,6]decyl, 3-methyl-3-tetracyclo[4.4.0.1.sup.2,5.1.sup.7,10]dodecyl, 3-ethyl-3-tetracyclo[4.4.0.1.sup.2,5.1.sup.7,10]dodecyl, 2-methyl-2-adamantyl, 2-ethyl-2-adamantyl, 1-methyl-3-oxo-1-cyclohexyl, 1-methyl-1-(tetrahydrofuran-2-yl)ethyl, 5-hydroxy-2-methyl-2-adamantyl, and 5-hydroxy-2-ethyl-2-adamantyl.

[0170] A group having the following formula (B2-1-1) is also suitable as the acid labile group. The group having formula (B2-1-1) is often used as the acid labile group. It is a good choice of the acid labile group that ensures to form a pattern having a relatively rectangular pattern-substrate interface in a consistent manner. An acetal structure is formed when X is a group having formula (B2-1-1).

##STR00437##

[0171] In formula (B2-1-1), R.sup.L1 is hydrogen or a C.sub.1-C.sub.10 saturated hydrocarbyl group. The saturated hydrocarbyl group may be straight, branched or cyclic.

[0172] A choice of R.sup.L1 may depend on the designed sensitivity of labile group to acid. For example, hydrogen or a group in which the carbon atom bonded to acetal carbon is tertiary is selected when the acid labile group is designed to ensure relatively high stability and to be decomposed with strong acid. Examples of R.sup.L1 bonded to acetal carbon via tertiary carbon include tert-butyl, tert-pentyl, and 1-adamantyl, but are not limited thereto. A straight alkyl group is selected when the acid labile group is designed to have relatively high reactivity and high sensitivity to pH changes. Although the choice varies with a particular combination of acid

generator and quencher in the resist composition, R.sup.L1 is preferably a group in which the carbon in bond with acetal carbon is secondary, when R.sup.L2 is a relatively large alkyl group substituted at the end and the acid labile group is designed to undergo a substantial change of solubility by decomposition. Examples of R.sup.L1 bonded to acetal carbon via secondary carbon include isopropyl, sec-butyl, cyclopentyl, and cyclohexyl, but are not limited thereto.

[0173] In formula (B2-1-1), R.sup.L2 is a C.sub.1-C.sub.30 hydrocarbyl group. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Some constituent —CH.sub.2— in the hydrocarbyl group may be replaced by a heteroatom such as oxygen or sulfur so that the group may contain an ether bond or sulfide bond. Illustrative are C.sub.1-C.sub.30 saturated hydrocarbyl groups and C.sub.6-C.sub.30 aryl groups. R.sup.L2 is preferably a C.sub.1-C.sub.6 hydrocarbyl group for acquiring a higher resolution in forming small-size patterns. When R.sup.L2 is a C.sub.1-C.sub.6 hydrocarbyl group, the alcohol created after a progress of acid-aided deprotection reaction is water soluble. Then, when a positive pattern is formed using an alkaline developer, the alcohol is dissolved in the developer so that defects remaining in the exposed region are minimized.

[0174] Preferred examples of the group having formula (B2-1-1) are given below, but not limited thereto. Herein R.sup.L1 is as defined above.

##STR00438## ##STR00439##

[0175] Another acid labile group which can be used herein is a phenolic hydroxy group whose hydrogen is substituted by —CH.sub.2COO— (tertiary saturated hydrocarbyl group). The tertiary saturated hydrocarbyl group may be the same as the foregoing tertiary saturated hydrocarbyl group used for the protection of a phenolic hydroxy group.

[0176] Another example of repeat unit B2 is a repeat unit having the following formula (B2-2), referred to as repeat unit B2-2. The repeat unit B2-2 which enables to increase the dissolution rate in the exposed region is a useful choice of the acid labile group-containing unit which affords satisfactory performance against line width variations during develop loading.

##STR00440##

[0177] In formula (B2-2), c1 is an integer of 0 to 2, c2 is an integer of 0 to 2, c3 is an integer of 0 to 5, and c4 is an integer of 0 to 2.

[0178] In formula (B2-2), R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl.

[0179] In formula (B2-2), R.sup.32 and R.sup.33 are each independently a C.sub.1-C.sub.10 hydrocarbyl group which may contain a heteroatom. R.sup.32 and R.sup.33 may bond together to form a ring with the carbon atom to which they are attached.

[0180] In formula (B2-2), R.sup.34 is each independently fluorine, C.sub.1-C.sub.5 fluorinated alkyl group or C.sub.1-C.sub.5 fluorinated alkoxy group.

[0181] In formula (B2-2), R.sup.35 is each independently a C.sub.1-C.sub.10 hydrocarbyl group which may contain a heteroatom.

[0182] In formula (B2-2), A.sup.3 is a single bond, phenylene group, naphthylene group, or *—C(=O)—O-A.sup.31-. A.sup.31 is a C.sub.1-C.sub.20 aliphatic hydrocarbylene group which may contain hydroxy, ether bond, ester bond or lactone ring, or phenylene or naphthylene group, and * is a point of attachment to the carbon atom in the backbone.

[0183] Preferred examples of the repeat unit B2-2 are shown below, but not limited thereto. Herein R.sup.A is as defined above.

##STR00441## ##STR00442##

[0184] The content of repeat units B2 is preferably 5 to 95 mol %, more preferably 20 to 80 mol % based on the overall repeat units of the polymer. Each of repeat units B2 may be of one type or a mixture of two or more types.

[0185] In a preferred embodiment, the polymer further comprises repeat units of at least one type selected from units having the formulae (B3), (B4) and (B5). These repeat units are simply referred to as repeat units B3, B4 and B5, respectively.

##STR00443##

[0186] In formulae (B3) and (B4), d is an integer of 0 to 6 and e is an integer of 0 to 4.

[0187] In formulae (B3) and (B4), R.sup.41 and R.sup.42 are each independently hydroxy, halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group. The saturated hydrocarbyl group, saturated hydrocarbyloxy group and saturated hydrocarbylcarbonyloxy group may be straight, branched or cyclic. When d is 2 or more, a plurality of R.sup.41 may be identical or different. When e is 2 or more, a plurality of R.sup.42 may be identical or different.

[0188] In formula (B5), f1 is 0 or 1. The subscript f2 is an integer of 0 to 2, and the corresponding structure represents a benzene skeleton when f2=0, a naphthalene skeleton when f2=1, and an anthracene skeleton when f2=2. The subscript f3 is an integer of 0 to 5. In case of f2=0, preferably f3 is 0, 1, 2 or 3. In case of f2=1 or 2, preferably f3 is 0, 1, 2, 3 or 4.

[0189] In formula (B5), R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl.

[0190] In formula (B5), R.sup.43 is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.2-C.sub.20 saturated hydrocarbylcarbonyloxy group, C.sub.2-C.sub.20 saturated hydrocarbyloxyhydrocarbyl group, C.sub.2-C.sub.20 saturated hydrocarbylthiohydrocarbyl group, halogen atom, nitro group, or cyano group. In case of f2=1 or 2, R.sup.43 may also be hydroxy. The saturated hydrocarbyl group, saturated hydrocarbyloxy group, saturated hydrocarbylcarbonyloxy group, saturated hydrocarbyloxyhydrocarbyl group, and saturated hydrocarbylthiohydrocarbyl group may be straight, branched or cyclic. When f3 is 2 or more, a plurality of R.sup.43 may be identical or different.

[0191] In formula (B5), A.sup.4 is a single bond or a C.sub.1-C.sub.10 saturated hydrocarbylene group in which some constituent —CH.sub.2— may be replaced by —O—. The saturated hydrocarbylene group may be straight, branched or cyclic. Examples thereof are as exemplified above for A.sup.1 in formula (B1).

[0192] When repeat units of at least one type selected from repeat units B3 to B5 are incorporated, better performance is obtained because not only the aromatic ring possesses etch resistance, but the cyclic structure incorporated into the backbone also exerts the effect of improving etch resistance and resistance to EB irradiation during pattern inspection step.

[0193] The content of repeat units B3 to B5 is preferably at least 5 mol % based on the overall repeat units of the polymer for obtaining the effect of improving etch resistance. Also, the content of repeat units B3 to B5 is preferably up to 25 mol %, more preferably up to 20 mol % based on the overall repeat units of the polymer. When the relevant units are free of functional groups or have a functional group other than hydroxy, their content of up to 25 mol % is preferred because the risk of forming development defects is eliminated. Each of the repeat units B3 to B5 may be of one type or a combination of plural types.

[0194] It is preferred that the polymer comprise repeat units B1, repeat units B2, and repeat units of at least one type selected from repeat units B3 to B5, because both etch resistance and high resolution are achievable. The total content of these repeat units is preferably at least 60 mol %, more preferably at least 70 mol %, even more preferably at least 80 mol %, most preferably at least 90 mol % based on the overall repeat units of the polymer.

[0195] In another preferred embodiment, the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B6), repeat units having the formula (B7), repeat units having the formula (B8), and repeat units having the formula (B9), shown below. Notably these repeat units are also referred to as repeat units B6 to B9.

##STR00444##

[0196] In formulae (B6) to (B9), R.sup.A is each independently hydrogen, fluorine, methyl or trifluoromethyl. X.sup.1 is a single bond or phenylene group. X.sup.2 is *.sup.1—C(=O)—O—X.sup.21—, *.sup.1—C(=O)—NH—X.sup.21—, or *.sup.1—O—X.sup.21—. X.sup.21 is a

C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, or divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, and *.sup.1 designates a point of attachment to X.sup.1. X.sup.3 is each independently a single bond, phenylene group, naphthylene group or *.sup.2—C(=O)—O—X.sup.31—. X.sup.31 is a C.sub.1-C.sub.10 aliphatic hydrocarbylene group, phenylene group, or naphthylene group, the aliphatic hydrocarbylene group may contain a hydroxy moiety, ether bond, ester bond or lactone ring, and *.sup.2 designates a point of attachment to the carbon atom in the backbone. X.sup.4 is each independently a single bond, *.sup.3—X.sup.41—C(=O)—O—, *.sup.3—C(=O)—NH—X.sup.41— or *.sup.3—O—X.sup.41. X.sup.41 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, and *.sup.3 designates a point of attachment to X.sup.3. X.sup.5 is each independently a single bond, *.sup.4—X.sup.15—C(=O)—O—, *.sup.4—C(=O)—NH—X.sup.5— or *.sup.4—O—X.sup.5—. X.sup.51 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, and *.sup.4 designates a point of attachment to X.sup.4. X.sup.6 is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *.sup.2—C(=O)—O—X.sup.61—, *.sup.2—C(=O)—N(H)—X.sup.61—, or *.sup.2—O—X.sup.61—. X.sup.61 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group, or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, and *.sup.2 designates a point of attachment to the carbon atom in the backbone.

[0197] The aliphatic hydrocarbylene group represented by X.sup.21, X.sup.31 and X.sup.61 may be straight, branched or cyclic. Examples thereof include alkanediyl groups such as methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,1-diyl, propane-1,2-diyl, propane-1,3-diyl, propane-2,2-diyl, butane-1,1-diyl, butane-1,2-diyl, butane-1,3-diyl, butane-2,3-diyl, butane-1,4-diyl, 1,1-dimethylethane-1,2-diyl, pentane-1,5-diyl, 2-methylbutane-1,2-diyl, and hexane-1,6-diyl; cycloalkanediyl groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanedyl, and cyclohexanediyl; and combinations thereof.

[0198] The hydrocarbylene group represented by X.sup.41 and X.sup.51 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are shown below, but not limited thereto.

##STR00445##

[0199] In formula (B6), R.sup.51 and R.sup.52 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.20 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, and tert-butyl; C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, and adamantyl; C.sub.2-C.sub.20 alkenyl groups such as vinyl, allyl, propenyl, butenyl, and hexenyl; C.sub.3-C.sub.20 cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C.sub.6-C.sub.20 aryl groups such as phenyl, naphthyl and thienyl; C.sub.7-C.sub.20 aralkyl groups such as benzyl, 1-phenylethyl and 2-phenylethyl, and combinations thereof. Inter alia, aryl groups are preferred. In the hydrocarbyl group, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen and some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—), or haloalkyl moiety.

[0200] R.sup.51 and R.sup.52 may bond together to form a ring with the sulfur atom to which they are attached. Examples of the ring are as exemplified above for the ring that R.sup.ct1 and R.sup.ct2 in formula (cation-1) form with the sulfur atom to which they are attached.

[0201] Examples of the cation in repeat units B6 are shown below, but not limited thereto. Herein

R.sup.A is as defined above.

##STR00446## ##STR00447## ##STR00448## ##STR00449## ##STR00450## ##STR00451##
##STR00452## ##STR00453##

[0202] In formula (B6), M.sup.— is a non-nucleophilic counter ion. Halide ions, sulfonate anions, imide anions, and methide anions are preferred. Examples of the non-nucleophilic counter ion include halide ions such as chloride and bromide ions; sulfonate anions, specifically fluoroalkylsulfonate ions such as triflate, 1,1,1-trifluoroethanesulfonate, and nonafluorobutanesulfonate, arylsulfonate ions such as tosylate, benzenesulfonate, 4-fluorobenzenesulfonate, and 1,2,3,4,5-pentafluorobenzenesulfonate, alkylsulfonate ions such as mesylate and butanesulfonate; imide ions such as bis(trifluoromethylsulfonyl)imide, bis(perfluoroethylsulfonyl)imide and bis(perfluorobutylsulfonyl)imide; and methide ions such as tris(trifluoromethylsulfonyl)methide and tris(perfluoroethylsulfonyl)methide.

[0203] Anions having the following formulae (B6-1) to (B6-4) are also useful as the non-nucleophilic counter ion.

##STR00454##

[0204] In formula (B6-1), R.sup.fa is fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as will be exemplified below for the hydrocarbyl group R.sup.fa1 in formula (B6-1-1).

[0205] Of the anions of formula (B6-1), an anion having the formula (B6-1-1) is preferred.

##STR00455##

[0206] In formula (B6-1-1), Q.sup.11 and Q.sup.12 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred for solvent solubility that at least one of Q.sup.11 and Q.sup.12 be trifluoromethyl. The subscript m is an integer of 0 to 4, preferably 1.

[0207] R.sup.fa1 is a C.sub.1-C.sub.35 hydrocarbyl group which may contain a heteroatom. As the heteroatom, oxygen, nitrogen, sulfur and halogen atoms are preferred, with oxygen being most preferred. Of the hydrocarbyl groups, those groups of 6 to 30 carbon atoms are preferred from the aspect of achieving a high resolution in forming patterns of small feature size. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.35 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl, neopentyl, hexyl, heptyl, 2-ethylhexyl, nonyl, undecyl, tridecyl, pentadecyl, heptadecyl, and icosyl; C.sub.3-C.sub.35 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, 1-adamantyl, 2-adamantyl, 1-adamantylmethyl, norbornyl, norbornylmethyl, tricyclodecyl, tetracyclododecyl, tetracyclododecylmethyl, and dicyclohexylmethyl; C.sub.2-C.sub.35 unsaturated aliphatic hydrocarbyl groups such as allyl and 3-cyclohexenyl; C.sub.6-C.sub.35 aryl groups such as phenyl, 1-naphthyl, 2-naphthyl and 9-fluorenyl; and C.sub.7-C.sub.35 aralkyl groups such as benzyl and diphenylmethyl, and combinations thereof.

[0208] In the foregoing hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, or some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, nitro, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Examples of the heteroatom-containing hydrocarbyl group include tetrahydrofuryl, methoxymethyl, ethoxymethyl, methylthiomethyl, acetamidomethyl, trifluoroethyl, (2-methoxyethoxy)methyl, acetoxymethyl, 2-carboxy-1-cyclohexyl, 2-oxopropyl, 4-oxo-1-adamantyl, and 3-oxocyclohexyl.

[0209] In formula (B6-1-1), L.sup.a1 is a single bond, ether bond, ester bond, sulfonate ester bond, carbonate bond or carbamate bond. From the aspect of synthesis, an ether bond or ester bond is

preferred, with the ester bond being more preferred.

[0210] Examples of the anion having formula (B6-1) are shown below, but not limited thereto. Herein Q.sup.11 is as defined above.

##STR00456## ##STR00457## ##STR00458## ##STR00459## ##STR00460## ##STR00461##
##STR00462## ##STR00463##

[0211] In formula (B6-2), R.sup.fb1 and R.sup.fb2 are each independently fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for R.sup.fa1 in formula (B6-1-1). Preferably R.sup.fb1 and R.sup.fb2 are fluorine or C.sub.1-C.sub.4 straight fluorinated alkyl groups. Also, R.sup.fb1 and R.sup.fb2 may bond together to form a ring with the linkage: —CF.sub.2—SO.sub.2—N—SO.sub.2—CF.sub.2— to which they are attached. It is preferred that a combination of R.sup.fb1 and R.sup.fb2 be a fluorinated ethylene or fluorinated propylene group.

[0212] In formula (B6-3), R.sup.fc1, R.sup.fc2 and Rf.sup.fc3 are each independently fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified for R.sup.fa1 in formula (B6-1-1). Preferably R.sup.fc1, R.sup.fc2 and Rf.sup.fc3 are fluorine or C.sub.1-C.sub.4 straight fluorinated alkyl groups. Also, R.sup.fc1 and R.sup.fc2 may bond together to form a ring with the linkage: —CF.sub.2—SO.sub.2—C.sup.—SO.sub.2—CF.sub.2— to which they are attached. It is preferred that a combination of R.sup.fc1 and R.sup.fc2 be a fluorinated ethylene or fluorinated propylene group.

[0213] In formula (B6-4), R.sup.fd is a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for R.sup.fa1 in formula (B6-1-1).

[0214] Examples of the anion having formula (B6-4) are shown below, but not limited thereto.

##STR00464## ##STR00465##

[0215] Anions having an iodized or brominated aromatic ring are also useful as the non-nucleophilic counter ion. These anions have the formula (B6-5).

##STR00466##

[0216] In formula (B6-5), p is an integer of 1 to 3, q is an integer of 1 to 5, r is an integer of 0 to 3, and q+r is from 1 to 5. Preferably, q is 1, 2 or 3, more preferably 2 or 3, and r is 0, 1 or 2.

[0217] In formula (B6-5), X.sup.BI is iodine or bromine. A plurality of X.sup.BI may be identical or different when p and/or q is 2 or more.

[0218] In formula (B6-5), L.sup.1 is a single bond, ether bond, ester bond, or a C.sub.1-C.sub.6 saturated hydrocarbylene group which may contain an ether bond or ester bond. The saturated hydrocarbylene group may be straight, branched or cyclic.

[0219] In formula (B6-5), L.sup.2 is a single bond or a C.sub.1-C.sub.20 divalent linking group when p=1, or a C.sub.1-C.sub.20 (p+1)-valent linking group when p=2 or 3. The linking group may contain an oxygen, sulfur or nitrogen atom.

[0220] In formula (B6-5), R.sup.fe is hydroxy, carboxy, fluorine, chlorine, bromine, amino group, or a C.sub.1-C.sub.20 hydrocarbyl, C.sub.1-C.sub.20 hydrocarbyloxy, C.sub.2-C.sub.20 hydrocarbylcarbonyl, C.sub.2-C.sub.20 hydrocarbyloxycarbonyl, C.sub.2-C.sub.20 hydrocarbylcarbonyloxy, or C.sub.1-C.sub.20 hydrocarbylsulfonyloxy group, which may contain fluorine, chlorine, bromine, hydroxy, amino or ether bond, or —N(R.sup.feA)(R.sup.feB), —N(R.sup.feC)—C(=O)—R.sup.feD or N(R.sup.feC)—C(=O)—O—R.sup.feD. R.sup.feA and R.sup.feB are each independently hydrogen or a C.sub.1-C.sub.6 saturated hydrocarbyl group. R.sup.feC is hydrogen, or a C.sub.1-C.sub.6 saturated hydrocarbyl group which may contain halogen, hydroxy, C.sub.1-C.sub.6 saturated hydrocarbyloxy, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyl or C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy moiety. R.sup.feD is a C.sub.1-C.sub.16 aliphatic hydrocarbyl group, C.sub.6-C.sub.12 aryl group or C.sub.7-C.sub.15

alkyl group, which may contain halogen, hydroxy, C.sub.1-C.sub.6 saturated hydrocarbyloxy, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyl or C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy moiety. The aliphatic hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. The hydrocarbyl, hydrocarbyloxy, hydrocarbylcarbonyl, hydrocarbyloxycarbonyl, hydrocarbylcarbonyloxy, and hydrocarbylsulfonyloxy groups may be straight, branched or cyclic. A plurality of R.sup.fe may be identical or different when p and/or r is 2 or more.

[0221] Of these, R.sup.fe is preferably hydroxy, —N(R.sup.feC)—C(=O)—R.sup.feD, —N(R.sup.feC)—C(=O)—O—R.sup.feD, fluorine, chlorine, bromine, methyl or methoxy.

[0222] In formula (B6-5), Rf.sup.11 to Rf.sup.14 are each independently hydrogen, fluorine or trifluoromethyl, at least one of Rf.sup.11 to Rf.sup.14 is fluorine or trifluoromethyl. Rf.sup.11 and Rf.sup.12, taken together, may form a carbonyl group. More preferably, both Rf.sup.13 and Rf.sup.14 are fluorine.

[0223] Examples of the anion having formula (B6-5) are shown below, but not limited thereto. X.sup.BI is as defined above.

##STR00467## ##STR00468## ##STR00469## ##STR00470## ##STR00471## ##STR00472##
##STR00473## ##STR00474## ##STR00475## ##STR00476## ##STR00477## ##STR00478##
##STR00479## ##STR00480## ##STR00481## ##STR00482## ##STR00483## ##STR00484##
##STR00485## ##STR00486## ##STR00487## ##STR00488## ##STR00489## ##STR00490##
##STR00491## ##STR00492## ##STR00493## ##STR00494## ##STR00495## ##STR00496##
##STR00497## ##STR00498## ##STR00499## ##STR00500## ##STR00501## ##STR00502##
##STR00503## ##STR00504## ##STR00505## ##STR00506##
##STR00507## ##STR00508## ##STR00509## ##STR00510## ##STR00511## ##STR00512##
##STR00513##
##STR00514## ##STR00515## ##STR00516## ##STR00517## ##STR00518##

[0224] Other useful examples of the non-nucleophilic counter ion include fluorobenzenesulfonic acid anions having an iodized aromatic ring bonded thereto as described in JP 6648726, anions having an acid-catalyzed decomposition mechanism as described in WO 2021/200056 and JP-A 2021-070692, anions having a cyclic ether group as described in JP-A 2018-180525 and JP-A 2021-035935, and anions as described in JP-A 2018-092159.

[0225] Further useful examples of the non-nucleophilic counter ion include fluorine-free bulky benzenesulfonic acid anions as described in JP-A 2006-276759, JP-A 2015-117200, JP-A 2016-065016, and JP-A 2019-202974; fluorine-free benzenesulfonic acid or alkylsulfonic acid anions having an iodized aromatic group bonded thereto as described in JP 6645464.

[0226] Also useful are bissulfonic acid anions as described in JP-A 2015-206932, sulfonamide or sulfonimide anions having sulfonic acid side and different side as described in WO 2020/158366, and anions having a sulfonic acid side and a carboxylic acid side as described in JP-A 2015-024989.

[0227] In formulae (B7) and (B8), L.sup.1 is a single bond, ether bond, ester bond, carbonyl, sulfonate ester bond, carbonate bond or carbamate bond. From the aspect of synthesis, an ether bond, ester bond or carbonyl is preferred, with the ester bond or carbonyl being more preferred.

[0228] In formula (B7), Rf.sup.1 and Rf.sup.2 are each independently fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred that both Rf.sup.1 and Rf.sup.2 be fluorine because the generated acid has a higher acid strength. Rf.sup.3 and Rf.sup.4 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred for solvent solubility that at least one of Rf.sup.3 and Rf.sup.4 be trifluoromethyl.

[0229] In formula (B8), Rf.sup.5 and Rf.sup.6 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is excluded that all Rf.sup.5 and Rf.sup.6 are hydrogen at the same time. It is preferred for solvent solubility that at least one of Rf.sup.5 and Rf.sup.6 be trifluoromethyl.

[0230] In formulae (B7) and (B8), g1 and g2 each are an integer of 0 to 3, preferably 1.

[0231] Examples of the anion in repeat unit B7 are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00519## ##STR00520## ##STR00521## ##STR00522## ##STR00523## ##STR00524##
##STR00525## ##STR00526## ##STR00527## ##STR00528## ##STR00529## ##STR00530##
##STR00531## ##STR00532## ##STR00533## ##STR00534## ##STR00535## ##STR00536##

[0232] Examples of the anion in repeat unit B8 are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00537## ##STR00538## ##STR00539## ##STR00540## ##STR00541## ##STR00542##
##STR00543## ##STR00544## ##STR00545## ##STR00546## ##STR00547## ##STR00548##
##STR00549## ##STR00550## ##STR00551## ##STR00552## ##STR00553## ##STR00554##

[0233] Examples of the anion in repeat unit B9 are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00555## ##STR00556##

[0234] In formulae (1B6) to (B9), A.sup.+ is an onium cation. Suitable onium cations include ammonium, sulfonium and iodonium cations, with the sulfonium and iodonium cations being preferred. Examples of the sulfonium cation are as exemplified above for the sulfonium cation having formula (cation-1). Examples of the iodonium cation are as exemplified above for the iodonium cation having formula (cation-2).

[0235] The repeat units B6 to B9 are capable of generating an acid upon exposure to high-energy radiation. It is believed that binding of the relevant units to a polymer enables to appropriately control acid diffusion and to form a pattern with reduced LER. Since the acid-generating unit is bound to a polymer, the phenomenon that acid volatilizes from the exposed region and re-deposits on the unexposed region during bake in vacuum is suppressed. This is effective for reducing LER and for suppressing profile degradation due to unwanted film thickness loss in the unexposed region.

[0236] Of repeat units B6 to B9, repeat units B7 to B9 are preferred for the processing of photomask blanks because an optimum acid strength is available for the suppression of acid diffusion and the design of an acid labile group on the polymer. The repeat units B7 and B8 are more preferred

[0237] When repeat units B6 to B9 are included, their content is preferably 0.1 to 30 mol %, more preferably 0.5 to 20 mol % based on the overall repeat units of the polymer. Each of repeat units B6 to B9 may be of one type or a combination of plural types.

[0238] The content of repeat units having an aromatic ring structure is preferably at least 65 mol %, more preferably at least 75 mol %, even more preferably at least 85 mol % based on the overall repeat units of the polymer. When the polymer does not contain repeat units B6 to B9, it is preferred that all units have an aromatic ring structure.

[0239] The polymer may further comprise (meth)acrylate units protected with an acid labile group or (meth)acrylate units having an adhesive group such as lactone structure or hydroxy group other than phenolic hydroxy as commonly used in the art. These repeat units are effective for fine adjustment of properties of a resist film, but not essential.

[0240] Examples of the (meth)acrylate unit having an adhesive group include repeat units having the following formulae (B10) to (B12), which are also referred to as repeat units B10 to B12. While these units do not exhibit acidity, they may be used as auxiliary units for providing adhesion to substrates or adjusting solubility.

##STR00557##

[0241] In formulae (B10) to (B12), R.sup.A is each independently hydrogen, fluorine, methyl or trifluoromethyl. R.sup.61 is —O— or methylene. R.sup.62 is hydrogen or hydroxy. R.sup.63 is a C.sub.1-C.sub.4 saturated hydrocarbyl group, and h is an integer of 0 to 3.

[0242] When repeat units B10 to B12 are included, their content is preferably 0 to 20 mol %, more

preferably 0 to 10 mol % based on the overall repeat units of the polymer. Each of repeat units B10 to B12 may be of one type or a combination of plural types.

[0243] The polymer may be synthesized by combining suitable monomers optionally protected with a protective group, copolymerizing them in the standard way, and effecting deprotection reaction if necessary. The copolymerization reaction is preferably radical or anionic polymerization though not limited thereto. For the polymerization reaction, reference may be made to JP-A 2004-115630, for example.

[0244] The polymer should preferably have a Mw of 1,000 to 50,000, and more preferably 2,000 to 20,000. A Mw of at least 1,000 eliminates the risk that pattern features are rounded at their top to invite degradations of resolution and LER. A Mw of up to 50,000 eliminates the risk that LER is degraded when a pattern with a line width of up to 100 nm is formed. As used herein, Mw is measured by GPC versus polystyrene standards using tetrahydrofuran (THF) or dimethylformamide (DMF) solvent.

[0245] The polymer preferably has a narrow molecular weight distribution or dispersity (Mw/Mn) of 1.0 to 2.0, more preferably 1.0 to 1.9. A polymer with such a narrow dispersity eliminates the risk that foreign particles are left on the pattern after development and the pattern profile is aggravated.

[0246] The base polymer is designed such that the dissolution rate in alkaline developer is preferably up to 10 nm/min, more preferably up to 7 nm/min, even more preferably up to 5 nm/min. In the advanced generation of lithography wherein the coating film on the substrate is in a thin film range of up to 100 nm, the influence of pattern film thickness loss during alkaline development becomes strong. When the polymer has an alkaline dissolution rate of greater than 10 nm/min, pattern collapse occurs, i.e., a small-size pattern cannot be formed. The problem becomes outstanding in the fabrication of photomasks requiring to be defectless and having a tendency of strong development process. It is noted that the dissolution rate of a base polymer in alkaline developer is computed by spin coating a 16.7 wt % solution of a polymer in propylene glycol monomethyl ether acetate (PGMEA) solvent onto a 8-inch silicon wafer, baking at 100° C. for 90 seconds to form a film of 1,000 nm thick, developing the film in a 2.38 wt % aqueous solution of tetramethylammonium hydroxide (TMAH) at 23° C. for 100 seconds, and measuring a loss of film thickness.

[0247] In addition to the polymer defined above, the base polymer (B) may contain another polymer. The other polymer may be any of prior art well-known base polymers used in resist compositions. The content of the other polymer is not particularly limited as long as the benefits of the invention are not impaired.

(C) Organic Solvent

[0248] The chemically amplified positive resist composition may comprise an organic solvent as component (C). The organic solvent used herein is not particularly limited as long as the components are soluble therein. Examples of the organic solvent are described in JP-A 2008-111103, paragraphs [0144] to [0145] (U.S. Pat. No. 7,537,880). Specifically, exemplary solvents include ketones such as cyclohexanone, cyclopentanone, methyl-2-n-pentyl ketone, and 2-heptanone; alcohols such as 3-methoxybutanol, 3-methyl-3-methoxybutanol, 1-methoxy-2-propanol, 1-ethoxy-2-propanol, and diacetone alcohol (DAA); ethers such as propylene glycol monomethyl ether (PGME), ethylene glycol monomethyl ether, propylene glycol monoethyl ether, ethylene glycol monoethyl ether, propylene glycol dimethyl ether, and diethylene glycol dimethyl ether; esters such as propylene glycol monomethyl ether acetate (PGMEA), propylene glycol monoethyl ether acetate, ethyl lactate (EL), ethyl pyruvate, butyl acetate, methyl 3-methoxypropionate, ethyl 3-ethoxypropionate, tert-butyl acetate, tert-butyl propionate, and propylene glycol mono-tert-butyl ether acetate; and lactones such as γ -butyrolactone (GBL), and mixtures thereof. Where an acid labile group of acetal form is used, a high boiling alcohol solvent such as diethylene glycol, propylene glycol, glycerol, 1,4-butanediol or 1,3-butanediol may be

added to accelerate deprotection reaction of acetal.

[0249] Of the above organic solvents, it is recommended to use 1-ethoxy-2-propanol, PGMEA, PGME, cyclohexanone, EL, GBL, and mixtures thereof.

[0250] In the resist composition, the organic solvent (C) is preferably used in an amount of 200 to 10,000 parts, more preferably 400 to 5,000 parts by weight per 80 parts by weight of the base polymer (B). The organic solvent may be used alone or in admixture.

(D) Fluorinated Polymer

[0251] The chemically amplified positive resist composition may further comprise a fluorinated polymer for the purposes of enhancing contrast, preventing chemical flare of acid upon exposure to high-energy radiation, preventing mixing of acid from an anti-charging film in the step of coating an anti-charging film-forming material on a resist film, and suppressing unexpected unnecessary pattern degradation. The fluorinated polymer contains repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3), and repeat units having the formula (D4), and may contain repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6). It is noted that repeat units having formulae (D1), (D2), (D3), (D4), (D5), and (D6) are also referred to as repeat units D1, D2, D3, D4, D5, and D6, respectively, hereinafter. Since the fluorinated polymer also has a surface-active function, it can prevent insoluble residues from re-depositing onto the substrate during the development step and is thus effective for preventing development defects.

##STR00558##

[0252] In formulae (D1) to (D6), j_1 is an integer of 1 to 3, j_2 is an integer meeting:

$0 \leq j_2 \leq 5 + 2(j_3) - j_1$, j_3 is 0 or 1, and k is an integer of 1 to 3. $R_{sup.C}$ is each independently hydrogen, fluorine, methyl or trifluoromethyl. $R_{sup.101}$, $R_{sup.102}$, $R_{sup.104}$ and $R_{sup.105}$ are each independently hydrogen or a $C_{sub.1-C_{sub.10}}$ saturated hydrocarbyl group. $R_{sup.103}$, $R_{sup.106}$, $R_{sup.107}$ and $R_{sup.108}$ are each independently hydrogen, a $C_{sub.1-C_{sub.15}}$ hydrocarbyl group or fluorinated hydrocarbyl group, or an acid labile group. An ether bond or carbonyl moiety may intervene in a carbon-carbon bond in the hydrocarbyl groups or fluorinated hydrocarbyl groups represented by $R_{sup.103}$, $R_{sup.106}$, $R_{sup.107}$ and $R_{sup.108}$. $R_{sup.109}$ is hydrogen or a $C_{sub.1-C_{sub.5}}$ straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond. $R_{sup.110}$ is a $C_{sub.1-C_{sub.5}}$ straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond. $R_{sup.111}$ is a $C_{sub.1-C_{sub.20}}$ saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine and some constituent $-CH_{sub.2}-$ may be replaced by an ester bond or ether bond. $Z_{sup.1}$ is a $C_{sub.1-C_{sub.20}}(k+1)$ -valent hydrocarbon group or $C_{sub.1-C_{sub.20}}(k+1)$ -valent fluorinated hydrocarbon group. $Z_{sup.2}$ is a single bond, $*-C(=O)-O-$ or $*-C(=O)-NH-$ wherein $*$ designates a point of attachment to the carbon atom in the backbone. $Z_{sup.3}$ is a single bond, $-O-$, $*-C(=O)-O-Z_{sup.31}-Z_{sup.32}-$ or $*-C(=O)-NH-Z_{sup.31}-Z_{sup.32}-$, wherein $Z_{sup.31}$ is a single bond or a $C_{sub.1-C_{sub.10}}$ saturated hydrocarbylene group, $Z_{sup.32}$ is a single bond, ester bond, ether bond or sulfonamide bond, and $*$ designates a point of attachment to the carbon atom in the backbone.

[0253] In formulae (D1) and (D2), the $C_{sub.1-C_{sub.10}}$ saturated hydrocarbyl group represented by $R_{sup.101}$, $R_{sup.102}$, $R_{sup.104}$ and $R_{sup.105}$ may be straight, branched or cyclic and examples thereof include $C_{sub.1-C_{sub.10}}$ alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-heptyl, n-octyl, n-nonyl, and n-decyl, and $C_{sub.3-C_{sub.10}}$ cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, adamantyl, and norbornyl. Inter alia, $C_{sub.1-C_{sub.6}}$ saturated hydrocarbyl groups are preferred.

[0254] In formulae (D1) to (D4), the $C_{sub.1-C_{sub.15}}$ hydrocarbyl group represented by

R.sup.103, R.sup.106, R.sup.107 and R.sup.108 may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.15 alkyl, C.sub.2-C.sub.15 alkenyl and C.sub.2-C.sub.15 alkynyl groups, with the alkyl groups being preferred. Suitable alkyl groups include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, n-pentyl, n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, n-undecyl, n-dodecyl, n-tridecyl, n-tetradecyl and n-pentadecyl. The fluorinated hydrocarbyl groups correspond to the foregoing hydrocarbyl groups in which some or all carbon-bonded hydrogen atoms are substituted by fluorine atoms.

[0255] In formula (D4), examples of the C.sub.1-C.sub.20 (k+1)-valent hydrocarbon group Z.sup.1 include the foregoing C.sub.1-C.sub.20 alkyl groups and C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups, with k number of hydrogen atoms being eliminated. Examples of the C.sub.1-C.sub.20 (k+1)-valent fluorinated hydrocarbon group Z.sup.1 include the foregoing (k+1)-valent hydrocarbon groups in which at least one hydrogen atom is substituted by fluorine.

[0256] Examples of the repeat units D1 to D4 are given below, but not limited thereto. Herein R.sup.B is as defined above.

##STR00559## ##STR00560## ##STR00561##

[0257] In formula (D5), examples of the C.sub.1-C.sub.5 hydrocarbyl groups R.sup.109 and R.sup.110 include alkyl, alkenyl and alkynyl groups, with the alkyl groups being preferred. Suitable alkyl groups include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, and n-pentyl. In these groups, a moiety containing a heteroatom such as oxygen, sulfur or nitrogen may intervene in a carbon-carbon bond.

[0258] In formula (D5), —OR.sup.109 is preferably a hydrophilic group. In this case, R.sup.109 is preferably hydrogen or a C.sub.1-C.sub.5 alkyl group in which oxygen intervenes in a carbon-carbon bond.

[0259] In formula (D5), Z.sup.2 is preferably *—C(=O)—O— or *—C(=O)—NH—. Also preferably R.sup.C is methyl. The inclusion of carbonyl in Z.sup.2 enhances the ability to trap the acid originating from the anti-charging film. A polymer wherein R.sup.C is methyl is a robust polymer having a high glass transition temperature (T_g) which is effective for suppressing acid diffusion. As a result, the resist film is improved in stability with time, and neither resolution nor pattern profile is degraded.

[0260] Examples of the repeat unit D5 are given below, but not limited thereto. Herein R.sup.C is as defined above.

##STR00562## ##STR00563## ##STR00564##

[0261] In formula (D6), the C.sub.1-C.sub.10 saturated hydrocarbylene group Z.sup.3 may be straight, branched or cyclic and examples thereof include methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,1-diyl, propane-1,2-diyl, propane-1,3-diyl, propane-2,2-diyl, butane-1,1-diyl, butane-1,2-diyl, butane-1,3-diyl, butane-2,3-diyl, butane-1,4-diyl, and 1,1-dimethylethane-1,2-diyl.

[0262] The C.sub.1-C.sub.20 saturated hydrocarbyl group having at least one hydrogen substituted by fluorine, represented by R.sup.111, may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.20 alkyl groups and C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups in which at least one hydrogen is substituted by fluorine.

[0263] Examples of the repeat unit D6 are given below, but not limited thereto. Herein R.sup.C is as defined above.

##STR00565## ##STR00566## ##STR00567## ##STR00568## ##STR00569## ##STR00570##

[0264] The content of repeat units D1 to D4 is preferably 15 to 95 mol %, more preferably 20 to 85 mol % based on the overall repeat units of the fluorinated polymer. The content of repeat unit D5 and/or D6 is preferably 5 to 85 mol %, more preferably 15 to 80 mol % based on the overall repeat units of the fluorinated polymer. Each of repeat units D1 to D6 may be used alone or in admixture.

[0265] The fluorinated polymer may comprise additional repeat units as well as the repeat units D1 to D6. Suitable additional repeat units include those described in U.S. Pat. No. 9,091,918 (JP-A 2014-177407, paragraphs [0046]-[0078]). When the fluorinated polymer comprises additional

repeat units, their content is preferably up to 50 mol % based on the overall repeat units.

[0266] The fluorinated polymer may be synthesized by combining suitable monomers optionally protected with a protective group, copolymerizing them in the standard way, and effecting deprotection reaction if necessary. The copolymerization reaction is preferably radical or anionic polymerization though not limited thereto. For the polymerization reaction, reference may be made to JP-A 2004-115630.

[0267] The fluorinated polymer should preferably have a Mw of 2,000 to 50,000, and more preferably 3,000 to 20,000. A fluorinated polymer with a Mw of less than 2,000 helps acid diffusion, degrading resolution and detracting from age stability. A polymer with too high Mw has a reduced solubility in solvent, with a risk of leaving coating defects. The fluorinated polymer preferably has a dispersity (Mw/Mn) of 1.0 to 2.2, more preferably 1.0 to 1.7.

[0268] In the resist composition, the fluorinated polymer (D) is preferably used in an amount of 0.01 to 30 parts by weight, more preferably 0.1 to 20 parts by weight per 80 parts by weight of the base polymer (B). The fluorinated polymer may be used alone or in admixture.

(E) Quencher

[0269] The chemically amplified positive resist composition preferably comprises a quencher as component (E). As used herein, the quencher refers to a compound capable of trapping an acid generated from the acid generator upon exposure. The quencher is effective for holding down the rate of diffusion of the acid (generated by the acid generator) in the resist film. Even when a substrate whose outermost surface is made of a chromium-containing material is used, the quencher is effective for suppressing the influence of the acid (generated in the resist film) on the chromium-containing material.

[0270] The quencher is typically selected from conventional basic compounds. Conventional basic compounds include primary, secondary, and tertiary aliphatic amines, mixed amines, aromatic amines, heterocyclic amines, nitrogen-containing compounds with carboxy group, nitrogen-containing compounds with sulfonyl group, nitrogen-containing compounds with hydroxy group, nitrogen-containing compounds with hydroxyphenyl group, alcoholic nitrogen-containing compounds, amide derivatives, imide derivatives, and carbamate derivatives. Also included are primary, secondary, and tertiary amine compounds, specifically amine compounds having a hydroxy, ether bond, ester bond, lactone ring, cyano, or sulfonate ester group as described in JP-A 2008-111103, paragraphs [0146]-[0164], and compounds having a carbamate group as described in JP 3790649. Inter alia, tris[2-(methoxymethoxy)ethyl]amine, tris[2-(methoxymethoxy)ethyl]amine-N-oxide, dibutylaminobenzoic acid, morpholine derivatives, and imidazole derivatives are preferred. Addition of a basic compound may be effective for further suppressing the diffusion rate of acid in the resist film or correcting the pattern profile.

[0271] Onium salts such as sulfonium, iodonium and ammonium salts of carboxylic acids which are not fluorinated at α -position as described in U.S. Pat. No. 8,795,942 (JP-A 2008-158339) may also be used as the quencher. While an α -fluorinated sulfonic acid, imide acid, and methide acid are necessary to deprotect the acid labile group, an α -non-fluorinated carboxylic acid is released by salt exchange with an α -non-fluorinated onium salt. The α -non-fluorinated carboxylic acid functions as a quencher because it does not induce substantial deprotection reaction.

[0272] Examples of the onium salt of α -non-fluorinated carboxylic acid include compounds having the formula (E1).

$R_{201}-CO_2-M_q$ (E1)

[0273] In formula (E1), R_{201} is hydrogen or a C₁-C₄₀ hydrocarbyl group which may contain a heteroatom, exclusive of the hydrocarbyl group in which the hydrogen bonded to the carbon atom at α -position of the carboxy group is substituted by fluorine or fluoroalkyl.

[0274] The hydrocarbyl group R_{201} may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C₁-C₄₀ alkyl groups such as methyl, ethyl, n-propyl,

isopropyl, n-butyl, sec-butyl, tert-butyl, n-pentyl, tert-pentyl, n-hexyl, n-octyl, 2-ethylhexyl, n-nonyl, n-decyl; C.sub.3-C.sub.40 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, cyclopentylmethyl, cyclopentylethyl, cyclopentylbutyl, cyclohexylmethyl, cyclohexylethyl, cyclohexylbutyl, norbornyl, tricyclo[5.2.1.0.sup.2,6]decyl, adamantyl, and adamantylmethyl; C.sub.2-C.sub.40 alkenyl groups such as vinyl, allyl, propenyl, butenyl and hexenyl; C.sub.3-C.sub.40 cyclic unsaturated aliphatic hydrocarbyl groups such as cyclohexenyl; C.sub.6-C.sub.40 aryl groups such as phenyl, naphthyl, alkylphenyl groups (e.g., 2-methylphenyl, 3-methylphenyl, 4-methylphenyl, 4-ethylphenyl, 4-tert-butylphenyl, 4-n-butylphenyl), di- or trialkylphenyl groups (e.g., 2,4-dimethylphenyl and 2,4,6-triisopropylphenyl), alkyl naphthyl groups (e.g., methyl naphthyl and ethyl naphthyl), dialkyl naphthyl groups (e.g., dimethyl naphthyl and diethyl naphthyl); and C.sub.7-C.sub.40 aralkyl groups such as benzyl, 1-phenylethyl and 2-phenylethyl.

[0275] In the hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy moiety, cyano moiety, carbonyl moiety, ether bond, thioether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—), or haloalkyl moiety. Suitable heteroatom-containing hydrocarbyl groups include heteroaryl groups such as thienyl; alkoxyphenyl groups such as 4-hydroxyphenyl, 4-methoxyphenyl, 3-methoxyphenyl, 2-methoxyphenyl, 4-ethoxyphenyl, 4-tert-butoxyphenyl, 3-tert-butoxyphenyl; alkoxy naphthyl groups such as methoxy naphthyl, ethoxy naphthyl, n-propoxy naphthyl and n-butoxy naphthyl; dialkoxy naphthyl groups such as dimethoxy naphthyl and diethoxy naphthyl; and aryloxoalkyl groups, typically 2-aryl-2-oxoethyl groups such as 2-phenyl-2-oxoethyl, 2-(1-naphthyl)-2-oxoethyl and 2-(2-naphthyl)-2-oxoethyl.

[0276] In formula (E1), Mq.sub.A.sup.+ is an onium cation. The onium cation is preferably selected from sulfonium, iodonium and ammonium cations, more preferably sulfonium and iodonium cations. Exemplary sulfonium cations are as exemplified above for the sulfonium cation having formula (cation-1). Exemplary iodonium cations are as exemplified above for the iodonium cation having formula (cation-2).

[0277] Examples of the anion in the onium salt having formula (E1) are shown below, but not limited thereto.

##STR00571## ##STR00572## ##STR00573## ##STR00574##

[0278] A sulfonium salt of iodized benzene ring-containing carboxylic acid having the formula (E2) is also useful as the quencher.

##STR00575##

[0279] In formula (E2), s is an integer of 1 to 5, t is an integer of 0 to 3, s+t is from 1 to 5, and u is an integer of 1 to 3.

[0280] In formula (E2), R.sup.211 is hydroxy, fluorine, chlorine, bromine, amino, nitro, cyano, or a C.sub.1-C.sub.6 saturated hydrocarbyl, C.sub.1-C.sub.6 saturated hydrocarbyloxy, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy or C.sub.1-C.sub.4 saturated hydrocarbylsulfonyloxy group, in which some or all hydrogen may be substituted by halogen, or —N(R.sup.211A)—C(=O)—R.sup.211B, or —N(R.sup.211A)—C(=O)—O—R.sup.211B. R.sup.211A is hydrogen or a C.sub.1-C.sub.6 saturated hydrocarbyl group. R.sup.211B is a C.sub.1-C.sub.6 saturated hydrocarbyl or C.sub.2-C.sub.8 unsaturated aliphatic hydrocarbyl group. A plurality of R.sup.211 may be identical or different when t and/or u is 2 or 3.

[0281] In formula (E2), L.sup.11 is a single bond or a C.sub.1-C.sub.20 (u+1)-valent linking group which may contain at least one moiety selected from ether bond, carbonyl moiety, ester bond, amide bond, sultone ring, lactam ring, carbonate bond, halogen, hydroxy moiety, and carboxy moiety. The saturated hydrocarbyl, saturated hydrocarbyloxy, saturated hydrocarbylcarbonyloxy, and saturated hydrocarbylsulfonyloxy groups may be straight, branched or cyclic.

[0282] In formula (E2), R.sup.212, R.sup.213 and R.sup.214 are each independently halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.20 alkyl, C.sub.2-C.sub.20 alkenyl, C.sub.6-C.sub.20 aryl, and C.sub.7-C.sub.20 aralkyl groups. In these groups, some or all hydrogen may be substituted by hydroxy, carboxy, halogen, oxo, cyano, nitro, sultone ring, sulfo, or sulfonium salt-containing moiety, or some —CH.sub.2— may be replaced by an ether bond, ester bond, carbonyl moiety, amide bond, carbonate bond or sulfonate ester bond. Also, R.sup.212 and R.sup.213 may bond together to form a ring with the sulfur atom to which they are attached.

[0283] Examples of the compound having formula (E2) include those described in U.S. Pat. No. 10,295,904 (JP-A 2017-219836). These compounds exert a sensitizing effect due to remarkable absorption and an acid diffusion-controlling effect.

[0284] A nitrogen-containing carboxylic acid salt compound having the formula (E3) is also useful as the quencher.

##STR00576##

[0285] In formula (E3), R.sup.221 to R.sup.224 are each independently hydrogen, -L.sup.12-CO.sub.2.sup.—, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. R.sup.221 and R.sup.222, R.sup.222 and R.sup.223, or R.sup.223 and R.sup.224 may bond together to form a ring with the carbon atom to which they are attached. L.sup.12 is a single bond or a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom. R.sup.225 is hydrogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom.

[0286] In formula (E3), the ring R.sup.r is a C.sub.2-C.sub.6 ring containing the carbon and nitrogen atoms in the formula, in which some or all of the carbon-bonded hydrogen atoms may be substituted by a C.sub.1-C.sub.20 hydrocarbyl group or -L.sup.12-CO.sub.2 and in which some carbon may be replaced by sulfur, oxygen or nitrogen. The ring may be alicyclic or aromatic and is preferably a 5- or 6-membered ring. Suitable rings include pyridine, pyrrole, pyrrolidine, piperidine, pyrazole, imidazoline, pyridazine, pyrimidine, pyrazine, imidazoline, oxazole, thiazole, morpholine, thiazine, and triazole rings.

[0287] The carboxylic onium salt having formula (E3) has at least one -L.sup.12-CO.sub.2—. That is, at least one of R.sup.221 to R.sup.224 is -L.sup.12-CO.sub.2—, and/or at least one of hydrogen atoms bonded to carbon atoms in the ring R.sup.r is substituted by -L.sup.12-CO.sub.2—.

[0288] In formula (E3), Mq.sub.B.sup.+ is a sulfonium, iodonium or ammonium cation, with the sulfonium cation being preferred. Examples of the sulfonium cation are as exemplified above for the sulfonium cation having formula (cation-1).

[0289] Examples of the anion in the compound having formula (E3) are shown below, but not limited thereto.

##STR00577## ##STR00578## ##STR00579## ##STR00580## ##STR00581## ##STR00582##

[0290] Weak acid betaine compounds are also useful as the quencher. Non-limiting examples thereof are shown below.

##STR00583##

[0291] Also useful are quenchers of polymer type as described in U.S. Pat. No. 7,598,016 (JP-A 2008-239918). The polymeric quencher segregates at the resist surface after coating and thus enhances the rectangularity of resist pattern. When a protective film is applied as is often the case in the immersion lithography, the polymeric quencher is also effective for preventing a film thickness loss of resist pattern or rounding of pattern top.

[0292] When used, the quencher (E) is preferably added in an amount of 0 to 50 parts, more preferably 0.1 to 40 parts by weight per 80 parts by weight of the base polymer (B). The quencher may be used alone or in admixture.

[0293] When the chemically amplified positive resist composition contains both the PAG (A) and the quencher (E), the weight ratio of the PAG to the quencher, (A)/(E) is preferably less than 3/1,

more preferably less than 2.5/1, even more preferably less than 2/1. As long as the weight ratio of the PAG to the quencher is in the range, the resist composition is able to fully suppress acid diffusion, leading to improved resolution and dimensional uniformity.

(F) Other Photoacid Generator

[0294] Besides the PAG in the form of the onium salt having formula (A) and the polymer-bound photoacid generator comprising units having formulae (B6) to (B9), the chemically amplified positive resist composition may further comprise another photoacid generator (PAG). The other PAG used herein may be any compound capable of generating an acid upon exposure to high-energy radiation. Suitable PAGs include sulfonium salts, iodonium salts, sulfonyldiazomethane, N-sulfonyloxyimide, and oxime-O-sulfonate acid generators.

[0295] Suitable other PAGs include nonafluorobutane sulfonate, partially fluorinated sulfonates described in JP-A 2012-189977, paragraphs [0247]-[0251], partially fluorinated sulfonates described in JP-A 2013-101271, paragraphs [0261]-[0265], and those described in JP-A 2008-111103, paragraphs [0122]-[0142] and JP-A 2010-215608, paragraphs [0080]-[0081]. Among others, arylsulfonate and alkanesulfonate type PAGs are preferred because they generate acids having an appropriate strength to deprotect the acid labile group in repeat unit B2.

[0296] The preferred PAGs are salt compounds having a sulfonium anion of the structure shown below.

##STR00584## ##STR00585## ##STR00586## ##STR00587## ##STR00588## ##STR00589##
##STR00590## ##STR00591## ##STR00592## ##STR00593## ##STR00594## ##STR00595##
##STR00596## ##STR00597## ##STR00598## ##STR00599##
##STR00600## ##STR00601## ##STR00602## ##STR00603## ##STR00604## ##STR00605##
##STR00606## ##STR00607## ##STR00608## ##STR00609## ##STR00610## ##STR00611##
##STR00612## ##STR00613## ##STR00614## ##STR00615## ##STR00616## ##STR00617##
##STR00618## ##STR00619## ##STR00620## ##STR00621## ##STR00622## ##STR00623##
##STR00624## ##STR00625## ##STR00626## ##STR00627## ##STR00628## ##STR00629##
##STR00630## ##STR00631## ##STR00632## ##STR00633## ##STR00634## ##STR00635##
##STR00636## ##STR00637## ##STR00638## ##STR00639## ##STR00640## ##STR00641##
##STR00642## ##STR00643## ##STR00644## ##STR00645## ##STR00646## ##STR00647##
##STR00648## ##STR00649## ##STR00650## ##STR00651## ##STR00652## ##STR00653##
##STR00654## ##STR00655## ##STR00656## ##STR00657## ##STR00658## ##STR00659##
##STR00660## ##STR00661## ##STR00662## ##STR00663## ##STR00664## ##STR00665##
##STR00666## ##STR00667## ##STR00668## ##STR00669## ##STR00670## ##STR00671##
##STR00672## ##STR00673## ##STR00674## ##STR00675## ##STR00676## ##STR00677##
##STR00678##
##STR00679## ##STR00680## ##STR00681## ##STR00682## ##STR00683## ##STR00684##
##STR00685## ##STR00686## ##STR00687## ##STR00688## ##STR00689## ##STR00690##
##STR00691## ##STR00692## ##STR00693## ##STR00694## ##STR00695## ##STR00696##
##STR00697## ##STR00698## ##STR00699## ##STR00700## ##STR00701##
##STR00702## ##STR00703## ##STR00704## ##STR00705## ##STR00706## ##STR00707##
##STR00708## ##STR00709## ##STR00710## ##STR00711## ##STR00712## ##STR00713##
##STR00714## ##STR00715##

[0297] Also preferred as the PAG is a salt compound containing an anion having the formula (F1).
##STR00716##

[0298] In formula (F1), m1 is 0 or 1, x is an integer of 1 to 3, y is an integer of 1 to 5, z is an integer of 0 to 3, and y+z is from 1 to 5.

[0299] In formula (F1), L.sup.21 is a single bond, ether bond, ester bond, sulfonate ester bond, carbonate bond or carbamate bond.

[0300] In formula (F1), L.sup.22 is an ether bond, ester bond, sulfonate ester bond, carbonate bond or carbamate bond.

[0301] In formula (F1), LD is a single bond or a C.sub.1-C.sub.20 hydrocarbylene group when $x=1$, and a C.sub.1-C.sub.20 ($x+1$)-valent hydrocarbon group when $x=2$ or 3. The hydrocarbylene group and ($x+1$)-valent hydrocarbon group may contain at least one moiety selected from ether bond, carbonyl moiety, ester bond, amide bond, sultone ring, lactam ring, carbonate bond, halogen, hydroxy moiety and carboxy moiety.

[0302] The C.sub.1-C.sub.20 hydrocarbylene group L.sup.D may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.20 alkanediyl groups such as methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, heptane-1,7-diyl, octane-1,8-diyl, nonane-1,9-diyl, decane-1,10-diyl, undecane-1,11-diyl, dodecane-1,12-diyl; C.sub.3-C.sub.20 cyclic saturated hydrocarbylene groups such as cyclopentanediy, cyclohexanediy, norbornanediy and adamantanediy; C.sub.2-C.sub.20 unsaturated aliphatic hydrocarbylene groups such as vinylene and propene-1,3-diyl; C.sub.6-C.sub.20 arylene groups such as phenylene and naphthylene; and combinations thereof. The C.sub.1-C.sub.20 ($x+1$)-valent hydrocarbon group L.sup.D may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include those exemplified above for the C.sub.1-C.sub.20 hydrocarbylene group, with one or two hydrogen atoms being eliminated.

[0303] In formula (F1), Rf.sup.21 and Rf.sup.22 are each independently hydrogen, fluorine or trifluoromethyl, at least one being fluorine or trifluoromethyl.

[0304] In formula (F1), R.sup.301 is hydroxy, carboxy, a C.sub.1-C.sub.6 saturated hydrocarbyl group, C.sub.1-C.sub.6 saturated hydrocarbyloxy group, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy group, fluorine, chlorine, bromine, $\text{—N(R.sup.301A)(R.sup.301B)—N(R.sup.301C)—C(=O)—R.sup.301D}$ or $\text{—N(R.sup.301C)—C(=O)—O—R.sup.301D}$. R.sup.301A and R.sup.301B are each independently hydrogen or a C.sub.1-C.sub.6 saturated hydrocarbyl group. R.sup.301C is hydrogen or a C.sub.1-C.sub.6 saturated hydrocarbyl group. R.sup.301D is a C.sub.2-C.sub.6 saturated hydrocarbyl group or C.sub.2-C.sub.8 unsaturated aliphatic hydrocarbyl group.

[0305] The C.sub.1-C.sub.6 saturated hydrocarbyl group represented by R.sup.301, R.sup.301A, R.sup.301B and R.sup.301C may be straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.6 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, and n-hexyl; and C.sub.3-C.sub.6 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclobutyl, cyclopentyl, and cyclohexyl. Examples of the saturated hydrocarbyl moiety in the C.sub.1-C.sub.6 saturated hydrocarbyloxy group represented by R.sup.301 are as exemplified above for the saturated hydrocarbyl group. Examples of the saturated hydrocarbyl moiety in the C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy group represented by R.sup.301 are as exemplified above for the C.sub.1-C.sub.6 saturated hydrocarbyl group, but of 1 to 5 carbon atoms.

[0306] The C.sub.2-C.sub.8 unsaturated aliphatic hydrocarbyl group represented by R.sup.301D may be straight, branched or cyclic and examples thereof include C.sub.2-C.sub.8 alkenyl groups such as vinyl, propenyl, butenyl, and hexenyl; C.sub.2-C.sub.8 alkynyl groups such as ethynyl, propynyl, and butynyl; and C.sub.3-C.sub.8 cyclic unsaturated aliphatic hydrocarbyl groups such as cyclohexenyl and norbornenyl.

[0307] In formula (F1), R.sup.302 is a C.sub.1-C.sub.20 saturated hydrocarbylene group or C.sub.6-C.sub.14 arylene group. Some or all of the hydrogen atoms in the saturated hydrocarbylene group may be substituted by halogen other than fluorine. Some or all of the hydrogen atoms in the arylene group may be substituted by a substituent selected from C.sub.1-C.sub.20 saturated hydrocarbyl groups, C.sub.1-C.sub.20 saturated hydrocarbyloxy groups, C.sub.6-C.sub.14 aryl groups, halogen, and hydroxy.

[0308] The C.sub.1-C.sub.20 hydrocarbylene group represented by R.sup.302 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the C.sub.1-C.sub.20 hydrocarbylene group L.sup.D.

[0309] Examples of the C.sub.6-C.sub.20 arylene group represented by R.sup.302 include phenylene, naphthylene, phenanthrenediyl, and anthracenediyl. The C.sub.1-C.sub.20 saturated hydrocarbyl moiety and hydrocarbyl moiety in the C.sub.1-C.sub.20 hydrocarbyloxy moiety, which are substituents on the arylene group, may be straight, branched or cyclic and examples thereof include C.sub.1-C.sub.20 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, heptadecyl, octadecyl, nonadecyl, and icosyl; and C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl and adamantyl. Examples of the C.sub.6-C.sub.14 arylene moiety which is a substituent on the arylene group include phenylene, naphthylene, phenanthrenediyl and anthracenediyl.

[0310] More preferably, the anion has the formula (F2).

##STR00717##

[0311] In formula (F2), x, y, z, L.sup.21, L.sup.D, and R.sup.301 are as defined above. The subscript m2 is an integer of 1 to 4. R.sup.302A is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.6-C.sub.14 aryl group, halogen or hydroxy group. When m2 is 2, 3 or 4, a plurality of R.sup.302A may be identical or different.

[0312] Examples of the anion having formula (F1) are shown below, but not limited thereto.

##STR00718## ##STR00719## ##STR00720## ##STR00721## ##STR00722## ##STR00723##
##STR00724## ##STR00725## ##STR00726## ##STR00727## ##STR00728## ##STR00729##
##STR00730## ##STR00731## ##STR00732## ##STR00733## ##STR00734## ##STR00735##
##STR00736## ##STR00737## ##STR00738## ##STR00739## ##STR00740## ##STR00741##
##STR00742## ##STR00743## ##STR00744##
##STR00745## ##STR00746## ##STR00747## ##STR00748## ##STR00749## ##STR00750##
##STR00751## ##STR00752## ##STR00753## ##STR00754## ##STR00755## ##STR00756##
##STR00757## ##STR00758## ##STR00759## ##STR00760## ##STR00761## ##STR00762##
##STR00763## ##STR00764##
##STR00765## ##STR00766## ##STR00767## ##STR00768## ##STR00769## ##STR00770##
##STR00771## ##STR00772## ##STR00773## ##STR00774## ##STR00775## ##STR00776##
##STR00777## ##STR00778## ##STR00779## ##STR00780## ##STR00781## ##STR00782##
##STR00783## ##STR00784## ##STR00785## ##STR00786## ##STR00787## ##STR00788##
##STR00789## ##STR00790## ##STR00791## ##STR00792## ##STR00793## ##STR00794##
##STR00795## ##STR00796## ##STR00797## ##STR00798## ##STR00799## ##STR00800##
##STR00801##

[0313] Preferred examples of the cation that pairs with the anion include sulfonium and iodonium cations. Examples of the sulfonium cation are as exemplified above for the sulfonium cation having formula (cation-1), but not limited thereto. Examples of the iodonium cation are as exemplified above for the iodonium cation having formula (cation-2), but not limited thereto.

[0314] The other PAG generates an acid having a pKa value of preferably -2.0 or larger, more preferably -1.0 or larger. The upper limit of pKa is preferably 2.0. Notably, the pKa value is computed using pKa DB in software ACD/Chemsketch ver: 9.04 of Advanced Chemistry Development Inc.

[0315] When the resist composition contains the other PAG (F), the amount of the PAG (F) used is preferably 1 to 10 parts, more preferably 1 to 5 parts by weight per 80 parts by weight of the base polymer (B). The inclusion of the other PAG provides for appropriate adjustment of the amount of acid generated in the exposed region and the degree of dissolution inhibition in the unexposed region. The other PAG may be used alone or in admixture.

[0316] In addition to the PAG (A), the preferred resist composition comprises the polymer-bound PAG containing repeat units having formulae (B6) to (B9) or the other PAG (F). This embodiment allows the base polymer in the resist film to undergo deprotection reaction to a full extent and is

successful in forming a pattern of satisfactory profile at an improved resolution.

(G) Surfactant

[0317] The resist composition may contain any conventional surfactants for facilitating to coat the composition to the substrate. A number of surfactants are known in the art as described in WO 2006/121096, JP-A 2008-102383, JP-A 2008-304590, JP-A 2004-115630, and JP-A 2005-008766, and any suitable one may be chosen therefrom.

[0318] When the resist composition contains the surfactant (G), the amount of the surfactant (G) added is preferably up to 2 parts by weight, more preferably up to 1 part by weight and preferably at least 0.01 part by weight per 80 parts by weight of the base polymer (B).

[Process]

[0319] Another embodiment of the invention is a pattern forming process comprising the steps of applying the chemically amplified positive resist composition defined above onto a substrate to form a resist film thereon, exposing the resist film to a pattern of high-energy radiation, and developing the exposed resist film in an alkaline developer.

[0320] The substrate used herein may be selected from, for example, substrates for IC fabrication, e.g., Si, SiO, SiO.sub.2, SiN, SiON, TiN, WSi, BPSG, SOG, and organic antireflective coating, and substrates for mask circuit fabrication, e.g., Cr, CrO, CrON, MoSi.sub.2, Si, SiO, and SiO.sub.2.

[0321] The resist composition is first applied onto a substrate by a suitable coating technique such as spin coating. The coating is prebaked on a hotplate preferably at a temperature of 60 to 150° C. for 1 to 20 minutes, more preferably at 80 to 140° C. for 1 to 10 minutes to form a resist film of 0.03 to 2 μm thick.

[0322] Then the resist film is exposed patternwise to high-energy radiation. Examples of the high-energy radiation include UV, deep UV, excimer laser radiation (typically, KrF and ArF), EB, EUV of wavelength 3 to 15 nm, X-ray, γ-ray, and synchrotron radiation.

[0323] On use of UV, deep UV, excimer laser radiation, EUV, X-ray, γ-ray, and synchrotron radiation, the resist film is exposed through a mask having the desired pattern, preferably in a dose of 1 to 300 mJ/cm.sup.2, more preferably 10 to 200 mJ/cm.sup.2. On use of EB, a pattern may be directly written preferably in a dose of 1 to 300 μC/cm.sup.2, more preferably 10 to 200 μC/cm.sup.2. The resist composition of the invention is particularly useful in the EUV and EB lithography processes.

[0324] The exposure may be performed by conventional lithography whereas the immersion lithography of holding a liquid, typically water between the resist film and the mask may be employed if desired. In the case of immersion lithography, a protective film which is insoluble in water may be formed on the resist film.

[0325] After the exposure, the resist film may be baked (PEB), for example, on a hotplate preferably at 60 to 150° C. for 1 to 20 minutes, and more preferably at 80 to 140° C. for 1 to 10 minutes.

[0326] Finally, development is carried out using as the developer an aqueous alkaline solution, such as a 0.1 to 5 wt %, preferably 2 to 3 wt %, aqueous solution of tetramethylammonium hydroxide (TMAH), this being done by a conventional method such as dip, puddle, or spray development for a period of 0.1 to 3 minutes, and preferably 0.5 to 2 minutes. In this way the exposed region of resist film is dissolved away, forming the desired pattern on the substrate.

[0327] The resist composition of the invention is advantageous particularly on use under the situation that requires high etching resistance, and a minimal change of pattern line width and minimal LER even when the time duration from exposure to PEB is prolonged. It is also advantageous for pattern formation on a substrate having a surface layer of material to which the resist pattern is less adherent with a likelihood of pattern stripping or pattern collapse, specifically a substrate having sputter deposited thereon a layer of metallic chromium or a chromium compound containing one or more light elements such as oxygen, nitrogen and carbon. The resist composition is particularly useful in forming a pattern on a photomask blank as the substrate.

EXAMPLES

[0328] Examples of the invention are given below by way of illustration and not by way of limitation. The abbreviation “pbw” is parts by weight. Analysis is made by IR and proton-NMR spectroscopy and time-of-flight mass spectrometry (TOF-MS). The analyzers used are shown below.

[0329] IR: NICOLET 6700 by Thermo Fisher Scientific Inc.

[0330] .sup.1H-NMR: ECA-500 by JEOL Ltd.

[0331] MALDI TOF-MS: S3000 by JEOL Ltd.

[1]Synthesis of Onium Salts

[Example 1-1]Synthesis of Onium Salt PAG-1

##STR00802##

(1) Synthesis of Intermediate In-1

[0332] In nitrogen atmosphere, a reactor was charged with 6.3 g of reactant SM-1, 9.8 g of reactant SM-2, 0.3 g of DMAP, and 50 g of methylene chloride and cooled in an ice bath. While the internal temperature of the reactor was kept below 20° C., 5.8 g of 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride in powder form was added. Thereafter, the reactor was allowed to reach room temperature, at which the reaction solution was aged for 12 hours. At the end of aging, water was added to quench the reaction, followed by ordinary aqueous work-up. The subsequent steps of distilling off the solvent and washing the residue with diisopropyl ether gave 13.9 g of Intermediate In-1 as oily matter (yield 92%).

(2) Synthesis of onium salt PAG-1

[0333] In nitrogen atmosphere, a reactor was charged with 13.9 g of Intermediate In-1, 7.8 g of reactant SM-3, 50 g of methylene chloride, and 30 g of water, which were stirred for 15 minutes. The organic layer was taken out, washed with water, and concentrated under reduced pressure. To the concentrate, 50 g of methyl isobutyl ketone was added, followed by azeotropic dewatering. Diisopropyl ether was then added for recrystallization, obtaining 15.5 g of the target onium salt, PAG-1 as white crystals (yield 92%).

[0334] PAG-1 was analyzed by IR spectroscopy and TOF-MS, with the data shown below. FIG. 1 is the .sup.1H-NMR/DMSO-d.sub.6 spectrum of PAG-1.

[0335] IR (D-ATR): ν =3447, 3064, 3022, 2971, 1751, 1626, 1594, 1583, 1507, 1464, 1442, 1379, 1368, 1322, 1272, 1226, 1204, 1159, 1132, 1113, 1064, 1040, 1021, 998, 905, 886, 832, 816, 762, 706, 684, 632, 601, 577, 534, 493 cm.^{sup.-1}

[0336] MALDI TOF-MS: [0337] positive M.^{sup.+} 277 (corresponding to C.sub.18H.sub.13OS.^{sup.+}) [0338] negative M.^{sup.-} 455 (corresponding to C.sub.2H.sub.19O.sub.5S.^{sup.-})

[Examples 1-2 to 1-10] Synthesis of Onium Salts PAG-2 to PAG-10

[0339] Onium salts PAG-2 to PAG-10 shown below were synthesized using corresponding reactants and well-known organic synthesis reactions.

##STR00803##

[2]Synthesis of Base Polymers

[Synthesis Examples 2-1 to 2-6]Synthesis of Base Polymers P-1 to P-6

[0340] Base polymers P-1 to P-6 were synthesized in a standard way by combining monomers, performing copolymerization reaction in a solvent, pouring the reaction solution to hexane for precipitation, washing the solid precipitate with hexane, isolation and drying. The polymer was analyzed for composition by .sup.1H-NMR and .sup.13C-NMR spectroscopy and for Mw and Mw/Mn by GPC versus polystyrene standards using THF solvent.

##STR00804##

[3] Preparation of Chemically Amplified Positive Resist Compositions

Examples 1-1 to 1-53 and Comparative Examples 1-1 to 1-28

[0341] A chemically amplified positive resist composition was prepared by dissolving selected

components in an organic solvent in accordance with the formulation shown in Tables 1 to 3, and filtering the solution through a nylon filter with a pore size of 5 nm and a UPE filter with a pore size of 1 nm. The organic solvent was a mixture of 940 pbw of PGMEA, 1,870 pbw of EL and 1,870 pbw of PGME.

TABLE-US-00001 TABLE 1 Resist Polymer 1 Polymer 2 Photoacid generator Quencher

Fluorinated polymer composition (pbw)	Polymer 1 (pbw)	Polymer 2 (pbw)	Photoacid generator (pbw)	Quencher (pbw)	Example
PAG-1 (12)	Q-1 (7.0)	FP-1 (1.5)	1-2	R-2 P-1 (80)	1-1 R-1 P-1 (80)
— PAG-2 (12)	Q-1 (7.0)	FP-1 (1.5)	1-4	R-4 P-1 (80)	1-3 R-3 P-1 (80)
R-5 P-1 (80)	— PAG-4 (12)	Q-1 (7.0)	FP-1 (1.5)	1-6 R-6 P-1 (80)	1-5 R-5 P-1 (80)
— PAG-6 (12)	Q-1 (7.0)	FP-1 (1.5)	1-8 R-8 P-1 (80)	— PAG-7 (12)	1-9 R-9 P-1 (80)
Q-1 (7.0)	FP-1 (1.5)	1-10 R-10 P-1 (80)	— PAG-9 (12)	Q-1 (7.0)	1-11 R-11 P-1 (80)
FP-1 (1.5)	1-12 R-12 P-1 (80)	— PAG-1 (8)	Q-2 (7.0)	FP-1 (1.5)	1-13 R-13 P-1 (80)
PAG-A (4)	1-14 R-14 P-1 (80)	— PAG-3 (12)	Q-3 (7.0)	FP-3 (1.5)	1-15 R-15 P-1 (80)
— PAG-6 (12)	Q-2 (7.0)	FP-4 (1.5)	1-16 R-16 P-2 (80)	— PAG-1 (12)	1-17 R-17 P-2 (80)
Q-1 (7.0)	FP-1 (1.5)	1-18 R-18 P-2 (80)	— PAG-3 (12)	Q-1 (7.0)	1-19 R-19 P-2 (80)
FP-1 (1.5)	1-20 R-20 P-2 (80)	— PAG-6 (12)	Q-1 (7.0)	FP-1 (1.5)	1-21 R-21 P-2 (80)
1-22 R-22 P-2 (80)	— PAG-10 (12)	Q-1 (7.0)	FP-1 (1.5)	1-23 R-23 P-2 (80)	1-24 R-24 P-2 (80)
— PAG-2 (8)	Q-3 (7.0)	FP-2 (1.5)	PAG-B (4)	.sup.	1-25 R-25 P-2 (80)
— PAG-6 (10)	Q-1 (6.8)	FP-1 (1.5)	PAG-B (2)	.sup.	1-26 R-26 P-3 (80)
— PAG-1 (12)	Q-1 (7.0)	FP-1 (1.5)	1-27 R-27 P-3 (80)	— PAG-2 (12)	1-28 R-28 P-3 (80)
Q-2 (6.8)	FP-1 (1.5)	1-29 R-29 P-3 (80)	— PAG-6 (12)	Q-4 (7.0)	1-30 R-30 P-3 (80)
FP-3 (1.5)	1-31 R-31 P-3 (80)	— PAG-10 (12)	Q-1 (7.0)	FP-1 (1.5)	

TABLE-US-00002 TABLE 2 Resist Polymer 1 Polymer 2 Photoacid generator Quencher

Fluorinated polymer composition (pbw)	Polymer 1 (pbw)	Polymer 2 (pbw)	Photoacid generator (pbw)	Quencher (pbw)	Example
PAG-1 (8)	Q-1 (7.0)	FP-4 (1.5)	PAG-A (4)	1-32 R-32 P-4 (80)	1-31 R-31 P-3 (80)
— PAG-2 (12)	Q-2 (6.8)	FP-1 (1.5)	1-34 R-34 P-4 (80)	— PAG-4 (12)	1-33 R-33 P-4 (80)
Q-1 (7.0)	FP-1 (1.5)	1-35 R-35 P-4 (80)	— PAG-5 (12)	Q-4 (7.0)	1-36 R-36 P-4 (80)
FP-3 (1.5)	1-37 R-37 P-4 (80)	— PAG-3 (8)	Q-1 (7.0)	FP-4 (1.5)	1-38 R-38 P-5 (80)
PAG-B (4).sup.	1-39 R-39 P-5 (80)	— PAG-2 (4)	Q-2 (7.2)	FP-1 (1.5)	1-40 R-40 P-5 (80)
— PAG-6 (4)	Q-1 (7.0)	FP-1 (1.5)	1-41 R-41 P-5 (80)	— PAG-8 (4)	1-42 R-42 P-5 (80)
Q-3 (7.0)	FP-1 (1.5)	1-43 R-43 P-5 (80)	— PAG-10 (4)	Q-4 (7.4)	1-44 R-44 P-6 (80)
FP-2 (1.5)	1-45 R-45 P-6 (80)	— PAG-1 (3)	Q-1 (7.2)	FP-1 (1.5)	1-46 R-46 P-6 (80)
PAG-A (1)	1-47 R-47 P-6 (80)	— PAG-7 (4)	Q-1 (7.0)	FP-1 (1.5)	1-48 R-48 P-6 (80)
1-49 R-49 P-6 (80)	— PAG-1 (3)	Q-1 (7.0)	FP-1 (1.5)	PAG-B (1).sup.	1-50 R-50 P-1 (40)
P-5 (40)	PAG-1 (4)	Q-1 (7.0)	FP-1 (1.5)	1-51 R-51 P-1 (40)	1-52 R-52 P-2 (40)
P-6 (40)	PAG-2 (4)	Q-1 (7.0)	FP-1 (1.5)	1-53 R-53 P-3 (40)	1-54 R-54 P-3 (40)
P-6 (40)	PAG-10 (4)	Q-3 (7.0)	FP-1 (1.5)		

TABLE-US-00003 TABLE 3 Resist Polymer 1 Polymer 2 Photoacid generator Quencher

Fluorinated polymer composition (pbw)	Polymer 1 (pbw)	Polymer 2 (pbw)	Photoacid generator (pbw)	Quencher (pbw)	Comparative
— cPAG-1 (12)	Q-1 (7.0)	FP-1 (1.5)	Example 1-2	CR-2 P-1 (80)	1-1 CR-1 P-1 (80)
— cPAG-2 (12)	Q-1 (7.0)	FP-1 (1.5)	1-4 CR-4 P-1 (80)	— cPAG-4 (12)	1-3 CR-3 P-1 (80)
Q-1 (7.0)	FP-1 (1.5)	1-5 CR-5 P-1 (80)	— cPAG-1 (8)	Q-2 (7.0)	1-6 CR-6 P-2 (80)
FP-1 (1.5)	1-7 CR-7 P-2 (80)	— cPAG-2 (12)	Q-3 (7.0)	FP-1 (1.5)	1-8 CR-8 P-2 (80)
1-9 CR-9 P-2 (80)	— cPAG-3 (12)	Q-2 (7.0)	FP-4 (1.5)	1-10 CR-10 P-3 (80)	1-11 CR-11 P-3 (80)
— cPAG-2 (8)	Q-3 (7.0)	FP-2 (1.5)	PAG-B (4)	1-12 CR-12 P-3 (80)	1-13 CR-13 P-3 (80)
— cPAG-1 (12)	Q-1 (7.0)	FP-1 (1.5)	1-14 CR-14 P-3 (80)	— cPAG-1 (8)	Q-1 (7.0)
FP-4 (1.5)	PAG-A (4)	1-15 CR-15 P-3 (80)	— cPAG-1 (12)	Q-1 (7.0)	FP-4 (1.5)

14 CR-14 P-4 (80) — cPAG-1 (12) Q-1 (7.0) FP-1 (1.5) 1-15 CR-15 P-4 (80) — cPAG-2 (12) Q-4 (7.0) FP-3 (1.5) 1-16 CR-16 P-4 (80) — cPAG-4 (12) Q-1 (7.0) FP-1 (1.5) 1-17 CR-17 P-4 (80) — cPAG-3 (8) Q-1 (7.0) FP-4 (1.5) PAG-B (4) 1-18 CR-18 P-5 (40) — cPAG-1 (4) Q-1 (7.0) FP-1 (1.5) 1-19 CR-19 P-5 (40) — cPAG-2 (4) Q-2 (7.2) FP-5 (1.5) 1-20 CR-20 P-5 (40) — cPAG-3 (4) Q-1 (7.0) FP-1 (1.5) 1-21 CR-21 P-5 (40) — cPAG-1 (3) Q-1 (7.2) FP-1 (1.5) PAG-A (1) 1-22 CR-22 P-6 (40) — cPAG-1 (4) Q-1 (7.0) FP-3 (1.5) 1-23 CR-23 P-6 (40) — cPAG-2 (4) Q-2 (7.0) FP-1 (1.5) 1-24 CR-24 P-6 (40) — PAG-1 (3) Q-1 (7.0) FP-1 (1.5) PAG-B (1) 1-25 CR-25 P-6 (40) — — Q-2 (7.0) FP-5 (1.5) 1-26 CR-26 P-1 (40) P-5 (40) cPAG-1 (4) Q-1 (7.0) FP-1 (1.5) 1-27 CR-27 P-2 (40) P-5 (40) cPAG-3 (4) Q-2 (7.0) FP-1 (1.5) 1-28 CR-28 P-1 (40) P-6 (40) cPAG-2 (2) Q-1 (7.0) FP-1 (1.5)

[0342] Comparative photoacid generators cPAG-1 to cPAG-4, photoacid generators PAG-A and PAG-B, Quenchers Q-1 to Q-4, and Fluorinated Polymers FP-1 to FP-5 in Tables 1 to 3 are identified below.

##STR00805## ##STR00806## ##STR00807## ##STR00808##

[4] EB Lithography Test

Examples 2-1 to 2-53 and Comparative Examples 2-1 to 2-28

[0343] A photomask blank of reflection type for an EUV lithography mask was furnished by starting with a low-coefficient-of-thermal-expansion glass substrate of 6 inches squares and depositing thereon a multilayer reflective film of 40 Mo/Si layers with a thickness of 284 nm, a Ru film of 3.5 nm thick as protective film, a TaN film of 70 nm thick as absorbing layer, and a CrN film of 6 nm thick as hard mask. Using a coater/developer system ACT-M (Tokyo Electron Ltd.), each of the resist compositions (R-1 to R-53, CR-1 to CR-28) was spin coated onto the photomask blank, and prebaked on a hotplate at 110° C. for 600 seconds to form a resist film of 80 nm thick. The thickness of the resist film was measured by an optical film thickness measurement system Nanospec (Nanometrics Inc.). Measurement was made at 81 points in the plane of the blank substrate excluding a peripheral band extending 10 mm inward from the blank periphery, and an average film thickness and a film thickness range were computed therefrom.

[0344] The resist film was exposed to EB using an EB writer system EBM-5000Plus (NuFlare Technology Inc., accelerating voltage 50 kV), then baked (PEB) at 110° C. for 600 seconds, and developed in a 2.38 wt % TMAH aqueous solution, thereby yielding a positive pattern.

[0345] The resist pattern was evaluated as follows. The patterned mask blank was observed under a top-down scanning electron microscope (TD-SEM). The optimum dose (Eop) was defined as the exposure dose ($\mu\text{C}/\text{cm}^2$) which provided a 1:1 resolution at the top and bottom of a 200-nm 1:1 line-and-space (LS) pattern. The resolution (or maximum IS resolution) was defined as the minimum size at the dose which provided a 9:1 resolution for an isolated space (IS) of 200 nm. The edge roughness (LER) of a 200-nm LS pattern was measured under SEM. The pattern was visually observed to judge whether or not the profile was rectangular. The results are shown in Tables 4 to 6.

TABLE-US-00004 TABLE 4 Resist Optimum dose Maximum IS LER Pattern composition ($\mu\text{C}/\text{cm}^2$) resolution (nm) (nm) profile Example 2-1 R-1 210 16 4.2 rectangular 2-2 R-2 200 17 4.4 rectangular 2-3 R-3 205 16 4.3 rectangular 2-4 R-4 210 18 4.3 rectangular 2-5 R-5 215 16 4.4 rectangular 2-6 R-6 210 18 4.3 rectangular 2-7 R-7 215 18 4.4 rectangular 2-8 R-8 220 17 4.2 rectangular 2-9 R-9 210 16 4.2 rectangular 2-10 R-10 215 18 4.4 rectangular 2-11 R-11 215 16 4.4 rectangular 2-12 R-12 210 17 4.4 rectangular 2-13 R-13 215 18 4.3 rectangular 2-14 R-14 210 16 4.2 rectangular 2-15 R-15 215 17 4.5 rectangular 2-16 R-16 210 17 4.2 rectangular 2-17 R-17 215 16 4.3 rectangular 2-18 R-18 210 18 4.5 rectangular 2-19 R-19 210 16 4.3 rectangular 2-20 R-20 200 17 4.5 rectangular 2-21 R-21 210 18 4.3 rectangular 2-22 R-22 215 16 4.3 rectangular 2-23 R-23 210 18 4.4 rectangular 2-24 R-24 215 17 4.4 rectangular 2-25 R-25 210 17 4.3 rectangular 2-26 R-26 215 16 4.2 rectangular 2-27 R-27 215 18 4.3 rectangular 2-28 R-28 205 17 4.2 rectangular 2-29 R-29 200 16 4.4 rectangular 2-30 R-30 205 18 4.4 rectangular

TABLE-US-00005 TABLE 5 Resist Optimum dose Maximum IS LER Pattern composition
 (μC/cm.sup.2) resolution (nm) (nm) profile Example 2-31 R-31 210 16 4.4 rectangular 2-32 R-32
 215 18 4.3 rectangular 2-33 R-33 210 16 4.3 rectangular 2-34 R-34 215 16 4.2 rectangular 2-35 R-
 35 220 17 4.4 rectangular 2-36 R-36 215 16 4.2 rectangular 2-37 R-37 210 17 4.3 rectangular 2-38
 R-38 205 18 4.3 rectangular 2-39 R-39 215 17 4.3 rectangular 2-40 R-40 210 18 4.3 rectangular 2-
 41 R-41 215 16 4.2 rectangular 2-42 R-42 205 18 4.4 rectangular 2-43 R-43 200 16 4.3 rectangular
 2-44 R-44 215 16 4.3 rectangular 2-45 R-45 215 17 4.2 rectangular 2-46 R-46 210 16 4.3
 rectangular 2-47 R-47 215 17 4.2 rectangular 2-48 R-48 205 18 4.3 rectangular 2-49 R-49 215 16
 4.4 rectangular 2-50 R-50 210 17 4.2 rectangular 2-51 R-51 215 16 4.3 rectangular 2-52 R-52 200
 17 4.3 rectangular 2-53 R-53 215 16 4.2 rectangular

TABLE-US-00006 TABLE 6 Resist Optimum dose Maximum IS LER Pattern composition
 (μC/cm.sup.2) resolution (nm) (nm) profile Comparative 2-1 CR-1 215 20 4.9 rectangular
 Example 2-2 CR-2 200 21 4.9 footing 2-3 CR-3 200 22 5.1 rectangular 2-4 CR-4 215 21
 4.8 rectangular 2-5 CR-5 205 20 5.1 footing 2-6 CR-6 210 23 5 rectangular 2-7 CR-7 200
 22 4.9 footing 2-8 CR-8 210 23 4.7 rectangular 2-9 CR-9 210 23 4.8 rectangular 2-10 CR-10
 210 21 4.9 rectangular 2-11 CR-11 215 20 5 rectangular 2-12 CR-12 210 22 4.9 footing 2-13 CR-
 13 210 23 4.7 rectangular 2-14 CR-14 210 24 4.8 rectangular 2-15 CR-15 215 23 5 footing 2-16
 CR-16 210 22 4.8 rectangular 2-17 CR-17 215 22 4.7 rectangular 2-18 CR-18 200 21 4.9 footing 2-
 19 CR-19 210 23 4.8 rectangular 2-20 CR-20 215 21 5 rectangular 2-21 CR-21 200 21 4.9
 rectangular 2-22 CR-22 205 23 5 footing 2-23 CR-23 215 24 4.8 rectangular 2-24 CR-24 200 24
 4.7 rectangular 2-25 CR-25 205 22 5 footing 2-26 CR-26 215 20 4.8 rectangular 2-27 CR-27 200
 20 4.9 footing 2-28 CR-28 205 20 4.7 rectangular

[0346] As is evident from Tables 4 to 6, the chemically amplified positive resist composition comprising the onium salt having formula (A) as a photoacid generator and the resist pattern forming process within the scope of the invention are effective in photolithography for the fabrication of semiconductor devices and the processing of photomask blanks of transmission and reflection types.

[0347] Japanese Patent Application No. 2024-031894 is incorporated herein by reference. Although some preferred embodiments have been described, many modifications and variations may be made thereto in light of the above teachings. It is therefore to be understood that the invention may be practiced otherwise than as specifically described without departing from the scope of the appended claims.

Claims

1. A chemically amplified positive resist composition comprising (A) a photoacid generator and (B) a base polymer containing a polymer adapted to be decomposed under the action of acid to increase its solubility in alkaline developer, said photoacid generator (A) being an onium salt having the formula (A): ##STR00809## wherein R.sup.1 to R.sup.12 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, one of R.sup.13 and R.sup.14 is a group having a partial structure having the formula (a) and the other is hydrogen, halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, at least two of R.sup.1 to R.sup.14 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and Z.sup.+ is an onium cation, ##STR00810## wherein m1 is 0 or 1, m2 is an integer of 0 to 4 when m1=0 and an integer of 0 to 6 when m1=1, m3 is an integer of 0 to 3 when m1=0 and an integer of 0 to 5 when m1=1, m2+m3 is from 0 to 4 when m1=0 and from 0 to 6 when m1=1, R.sup.F is fluorine, a C.sub.1-C.sub.6 fluorinated alkyl group, C.sub.1-C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group, R.sup.15 is a C.sub.1-C.sub.2M hydrocarbyl group which may contain a heteroatom, a plurality of R.sup.15 may bond together to form a ring with the carbon

atoms to which they are attached when m_3 is 2 or more, $L_{sup.A}$ and $L_{sup.B}$ are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and $X_{sup.L}$ is a single bond or a $C_{sub.1}-C_{sub.40}$ hydrocarbylene group which may contain a heteroatom, said polymer comprising repeat units having the formula (B1): $##STR00811##$ wherein a_1 is 0 or 1, a_2 is 0, 1 or 2, a_3 is an integer meeting $0 \leq a_3 \leq 5 + 2(a_2) - a_4$, a_4 is 1, 2 or 3, $R_{sup.A}$ is hydrogen, fluorine, methyl or trifluoromethyl, $R_{sup.21}$ is halogen, an optionally halogenated $C_{sub.1}-C_{sub.6}$ saturated hydrocarbyl group, optionally halogenated $C_{sub.1}-C_{sub.6}$ saturated hydrocarbyloxy group, or optionally halogenated $C_{sub.2}-C_{sub.8}$ saturated hydrocarbylcarbonyloxy group, and $A_{sup.1}$ is a single bond or $C_{sub.1}-C_{sub.10}$ saturated hydrocarbylene group in which some $-CH_{sub.2}-$ may be replaced by $-O-$.

2. The resist composition of claim 1 wherein the onium salt has the formula (A1): $##STR00812##$ wherein $R_{sup.1}$ to $R_{sup.13}$, $R_{sup.15}$, $L_{sup.A}$, $X_{sup.L}$, $R_{sup.F}$, m_1 to m_3 , and $Z_{sup.+}$ are as defined above.

3. The resist composition of claim 2 wherein the onium salt has the formula (A2): $##STR00813##$ wherein $R_{sup.5}$, $R_{sup.10}$ to $R_{sup.13}$, $R_{sup.15}$, $L_{sup.A}$, $X_{sup.L}$, $R_{sup.F}$, m_1 to m_3 , and $Z_{sup.+}$ are as defined above, m_4 and m_5 are each independently an integer of 0 to 4, $R_{sup.16}$ and $R_{sup.17}$ are each independently hydrogen, halogen, or a $C_{sub.1}-C_{sub.20}$ hydrocarbyl group which may contain a heteroatom; when m_4 is 2 or more, a plurality of $R_{sup.16}$ may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m_5 is 2 or more, a plurality of $R_{sup.17}$ may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

4. The resist composition of claim 1 wherein $Z_{sup.+}$ is a sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2): $##STR00814##$ wherein $R_{sup.ct1}$ to $R_{sup.ct5}$ are each independently halogen or a $C_{sub.1}-C_{sub.30}$ hydrocarbyl group which may contain a heteroatom, $R_{sup.ct1}$ and $R_{sup.ct2}$ may bond together to form a ring with the sulfur atom to which they are attached.

5. The resist composition of claim 1 wherein the polymer further comprises repeat units having the formula (B2-1): $##STR00815##$ wherein $R_{sup.A}$ is hydrogen, fluorine, methyl or trifluoromethyl, b_1 is 0 or 1, b_2 is 0, 1 or 2, b_3 is an integer meeting $0 \leq b_3 \leq 5 + 2(b_2) - b_4$, b_4 is 1, 2 or 3, b_5 is 0 or 1, $R_{sup.31}$ is halogen, an optionally halogenated $C_{sub.1}-C_{sub.6}$ saturated hydrocarbyl group, optionally halogenated $C_{sub.1}-C_{sub.6}$ saturated hydrocarbyloxy group, or optionally halogenated $C_{sub.2}-C_{sub.8}$ saturated hydrocarbylcarbonyloxy group, $A_{sup.2}$ is a single bond or $C_{sub.1}-C_{sub.10}$ saturated hydrocarbylene group in which some $-CH_{sub.2}-$ may be replaced by $-O-$, X is an acid labile group when $b_4=1$, and X is hydrogen or an acid labile group, at least one being an acid labile group, when $b_4=2$ or 3.

6. The resist composition of claim 1 wherein the polymer further comprises repeat units having the formula (B2-2): $##STR00816##$ wherein c_1 is 0, 1 or 2, c_2 is 0, 1 or 2, c_3 is an integer of 0 to 5, c_4 is 0, 1 or 2, $R_{sup.A}$ is hydrogen, fluorine, methyl or trifluoromethyl, $R_{sup.32}$ and $R_{sup.33}$ are each independently a $C_{sub.1}-C_{sub.10}$ hydrocarbyl group which may contain a heteroatom, $R_{sup.32}$ and $R_{sup.33}$ may bond together to form a ring with the carbon atom to which they are attached, $R_{sup.34}$ is each independently fluorine, a $C_{sub.1}-C_{sub.5}$ fluorinated alkyl group or $C_{sub.1}-C_{sub.5}$ fluorinated alkoxy group, $R_{sup.35}$ is each independently a $C_{sub.1}-C_{sub.10}$ hydrocarbyl group which may contain a heteroatom, and $A_{sup.3}$ is a single bond, phenylene, naphthylene or $*-C(=O)-O-A_{sup.31}$, $A_{sup.31}$ is a $C_{sub.1}-C_{sub.20}$ aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, a phenylene group or a naphthylene group, $*$ designates a point of attachment to the carbon atom in the backbone.

7. The resist composition of claim 1 wherein the polymer further comprises repeat units of at least

one type selected from repeat units having the formula (B3), repeat units having the formula (B4), and repeat units having the formula (B5): ##STR00817## wherein d is an integer of 0 to 6, e is an integer of 0 to 4, f1 is 0 or 1, f2 is 0, 1 or 2, f3 is an integer of 0 to 5, R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, R.sup.41 and R.sup.42 are each independently hydroxy, halogen, an optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyl group, optionally halogenated C.sub.1-C.sub.6 saturated hydrocarbyloxy group, or optionally halogenated C.sub.2-C.sub.8 saturated hydrocarbylcarbonyloxy group, R.sup.43 is a C.sub.1-C.sub.20 saturated hydrocarbyl group, C.sub.1-C.sub.20 saturated hydrocarbyloxy group, C.sub.2-C.sub.20 saturated hydrocarbylcarbonyloxy group, C.sub.2-C.sub.20 saturated hydrocarbyloxyhydrocarbyl group, C.sub.2-C.sub.20 saturated hydrocarbylthiohydrocarbyl group, halogen, nitro, or cyano, R.sup.43 may also be hydroxy when f2=1 or 2, and A.sup.4 is a single bond or C.sub.1-C.sub.10 saturated hydrocarbylene group in which some —CH.sub.2— may be replaced by —O—.

8. The resist composition of claim 1 wherein the polymer further comprises repeat units of at least one type selected from repeat units having the formula (B6), repeat units having the formula (B7), repeat units having the formula (B8), and repeat units having the formula (B9): ##STR00818## wherein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, X.sup.1 is a single bond or phenylene group, X.sup.2 is *.sup.1—C(=O)—O—X.sup.21, *.sup.1—C(=O)—NH—X.sup.21, or *.sup.1—O—X.sup.21—, X.sup.21 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group or a divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, *.sup.1 is a point of attachment to X.sup.1, X.sup.3 is each independently a single bond, phenylene, naphthylene, or *.sup.2—C(=O)—O—X.sup.31—, X.sup.31 is a C.sub.1-C.sub.10 aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, or phenylene or naphthylene group, *.sup.2 is a point of attachment to the carbon atom in the backbone, X.sup.4 is each independently a single bond, *.sup.3—X.sup.41—C(=O)—O—, *.sup.3—C(=O)—NH—X.sup.41—, or *.sup.3—O—X.sup.41—, X.sup.41 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, *.sup.3 is a point of attachment to X.sup.3, X.sup.5 is each independently a single bond, *.sup.4—X.sup.51—C(=O)—O—, *.sup.4—C(=O)—NH—X.sup.51—, or *.sup.4—O—X.sup.51—, X.sup.51 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, *.sup.4 is a point of attachment to X.sup.4, X.sup.6 is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *.sup.2—C(=O)—O—X.sup.61—, *.sup.2—C(=O)—N(H)—X.sup.61—, or *.sup.2—O—X.sup.61—, X.sup.61 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, *.sup.2 is a point of attachment to the carbon atom in the backbone, R.sup.51 and R.sup.52 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.51 and R.sup.52 may bond together to form a ring with the sulfur atom to which they are attached, L.sup.1 is a single bond, ether bond, ester bond, carbonyl group, sulfonate ester bond, carbonate bond or carbamate bond, Rf.sup.1 and Rf.sup.2 are each independently fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, Rf.sup.3 and Rf.sup.4 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, Rf.sup.5 and Rf.sup.6 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, excluding that all Rf.sup.5 and Rf.sup.6 are hydrogen at the same time, M.sup.— is a non-nucleophilic counter ion, A.sup.+ is an onium cation, and g1 and g2 are each independently an integer of 0 to 3.

9. The resist composition of claim 1 wherein repeat units having an aromatic ring structure account for at least 60 mol % of the overall repeat units of the polymer in the base polymer.

10. The resist composition of claim 1, further comprising (C) an organic solvent.

11. The resist composition of claim 1, further comprising (D) a fluorinated polymer comprising

repeat units of at least one type selected from repeat units having the formula (D1), repeat units having the formula (D2), repeat units having the formula (D3) and repeat units having the formula (D4) and optionally repeat units of at least one type selected from repeat units having the formula (D5) and repeat units having the formula (D6): ##STR00819## wherein j₁ is an integer of 1 to 3, j₂ is an integer meeting 0≤j₂≤5+2(j₃)-j₁, j₃ is 0 or 1, k is an integer of 1 to 3, R^{sup}.B is each independently hydrogen, fluorine, methyl or trifluoromethyl, R^{sup}.C is each independently hydrogen or methyl, R^{sup}.101, R^{sup}.102, R^{sup}.104 and R^{sup}.105 are each independently hydrogen or a C_{sub}.1-C_{sub}.10 saturated hydrocarbyl group, R^{sup}.103, R^{sup}.106, R^{sup}.107 and R^{sup}.108 are each independently hydrogen, a C_{sub}.1-C_{sub}.15 hydrocarbyl group, C_{sub}.1-C_{sub}.15 fluorinated hydrocarbyl group, or acid labile group, and when R^{sup}.103, R^{sup}.106, R^{sup}.107 and R^{sup}.108 each are a hydrocarbyl or fluorinated hydrocarbyl group, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond, R^{sup}.109 is hydrogen or a C_{sub}.1-C_{sub}.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, R^{sup}.110 is a C_{sub}.1-C_{sub}.5 straight or branched hydrocarbyl group in which a heteroatom-containing moiety may intervene in a carbon-carbon bond, R^{sup}.111 is a C_{sub}.1-C_{sub}.20 saturated hydrocarbyl group in which at least one hydrogen is substituted by fluorine, and in which some constituent —CH_{sub}.2— may be replaced by an ester bond or ether bond, Z^{sup}.1 is a C_{sub}.1-C_{sub}.20 (k+1)-valent hydrocarbon group or C_{sub}.1-C_{sub}.20 (k+1)-valent fluorinated hydrocarbon group, Z^{sup}.2 is a single bond, *—C(=O)—O— or *—C(=O)—NH—, * designates a point of attachment to the carbon atom in the backbone, Z^{sup}.3 is a single bond, —O—, *—C(=O)=O—Z^{sup}.31-Z^{sup}.32— or *—C(=O)—NH—Z^{sup}.31-Z^{sup}.32—, Z^{sup}.31 is a single bond or C_{sub}.1-C_{sub}.10 saturated hydrocarbylene group, Z^{sup}.32 is a single bond, ester bond, ether bond, or sulfonamide bond, and * designates a point of attachment to the carbon atom in the backbone.

12. The resist composition of claim 1, further comprising (E) a quencher.

13. The resist composition of claim 12 wherein the photoacid generator (A) and the quencher (E) are present in a weight ratio of less than 3/1.

14. The resist composition of claim 1, further comprising a photoacid generator other than the photoacid generator in the form of the onium salt having formula (A).

15. A resist pattern forming process comprising the steps of: applying the chemically amplified positive resist composition of claim 1 onto a substrate to form a resist film thereon, exposing the resist film patternwise to high-energy radiation, and developing the exposed resist film in an alkaline developer.

16. The process of claim 15 wherein the high-energy radiation is EUV of wavelength 3 to 15 nm or EB.

17. The process of claim 15 wherein the substrate has the outermost surface of a chromium-containing material.

18. The process of claim 15 wherein the substrate is a photomask blank.
